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Fluid, Electrolyte, and Acid-Base Disorders in Children

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CHAPTER OUTLINE

The pathophysiology of fluid and electrolyte disorders and the principles for their treatment are the same, irrespective of age. Yet, there are unique aspects to managing a child, especially in the first year of life:

- Body composition with respect to water and its distribution between extracellular and intracellular compartment changes substantially with age. In a 23-week premature baby, the relative size of the extracellular compartment is about three times larger, at 60% of body weight compared with an adult, where it is roughly 20%.[1,2]
- Similarly, the glomerular filtration rate (GFR), even when corrected for surface area, is much lower at 28 weeks' gestation, at around 10 mL/min/1.73 m^2 in the first week of life and increases to only around 15 mL/min/1.73 m^2 by 4 weeks of life.[3]
- Urinary concentrating ability is severely impaired in the neonate and reaches full potential only at around 1 year of age.[4,5]
- Childhood is especially vulnerable because of the need to facilitate growth and development. Disorders of fluid, electrolyte, and acid-base homeostasis can severely affect this, leading to rickets and stunted growth, and meticulous monitoring is required to adjust treatment to the needs of the growing body.

In this chapter, disorders of fluid, electrolyte, and acid-base homeostasis will be discussed, with particular attention to these specific aspects.

SODIUM AND WATER DISORDERS

NORMAL METABOLISM OF WATER AND SODIUM SPECIFIC TO PEDIATRICS

The most abundant constituent of the human body is water, but the relative amount changes with age. In newborns and infants, total body water (TBW) constitutes a greater proportion of body weight than in adults, approximately 90% at 23 weeks' gestation to 70% in term infants, and 65% at 12 months of age, compared with about 60% in an adult.[1,2]

The TBW is comprised of the intracellular fluid (ICF) and extracellular fluid (ECF), which may in turn be subdivided into intravascular and interstitial fluids. The age-dependent changes in TBW that occur during infancy and childhood are mostly attributable to variations in ECF because the ICF proportion remains essentially constant after the first year of life, at roughly 40% of the body weight. In contrast, the ECF decreases rapidly during the embryonic development stage, during the first year of life, and more slowly from 1 year of age onward.

In addition to the internal distribution of water, water and electrolyte metabolism depend on the external balance between intake and output. Children must keep a positive balance to allow body growth. A young infant gains approximately 30 g of weight/day, which includes the physiologic retention of 20 mL/day of water and 2 mEq/day of sodium. On the other hand, the ratio of surface area to weight is higher in infants than in adults, and the skin is more permeable in infants, probably due to the higher expression of water channels (aquaporins).[6,7] This means that water loss is proportionately much greater in infants than in adults, especially in children with fever.

HYPEROSMOLALITY AND HYPERNATREMIA: PATHOGENESIS AND CLASSIFICATION

Hyperosmolality results from a deficiency of water relative to solutes in the ECF. The vast majority of situations leading to hyperosmolality are attributable to losses of body water in excess of body solutes caused by insufficient water intake, excessive water excretion, or both, although a minority of cases can occur as a result of excessive total body sodium loading.[8] A list of the causal mechanisms of hyperosmolar disorders is summarized in Table 73.1 and demonstrates the specific vulnerability of young infants for developing hypernatremia. The conjunction of several factors, such as lack of voluntary access to water, high physiologic insensible water loss because of a large body surface in relation to body weight, occurrence of acute diarrhea frequently associated with fever, vomiting, and intolerance of oral fluid intake all contribute to this potentially dangerous vulnerability.

DEHYDRATION

It should be noted that in pediatrics the term "dehydration" is often used as the most general term, encompassing the different forms of net fluid loss, including forms in which the following are found: (1) water loss exceeds solute loss, leaving the child hypertonic; (2) water and solute loss are proportionate, leaving the child normotonic; and (3) solute loss exceeds water loss, leaving the child hypotonic. Although not as strictly rigorous as the term *net fluid volume depletion*, the term *dehydration* will be retained in this chapter to be consistent with terminology used in everyday pediatrics.

Hypernatremic Dehydration

Dehydration may occur as a result of inappropriately low intake (inadequate supply, anorexia, coma, fluid restriction) or excessive loss of fluids by gastrointestinal (GI; vomiting, diarrhea, fistula, or drains), renal (polyuric states), and/or cutaneous (heat, cystic fibrosis, burns, inflammatory skin disease) routes. Hypernatremic dehydration results from insufficient water intake or net loss of water in excess of solutes (see Table 73.1; i.e., the net fluid loss is hypotonic). In hypernatremic dehydration, water shifts from the ICF to the ECF, driven by an osmotic gradient. Thus, a certain degree of extracellular and intracellular volume loss coexists. In view of the fact that many of the most commonly assessed clinical manifestations are proportional to the reduction of ECF volume, the manifestations of dehydration will be less intense for the same degree of water loss in the hypernatremic dehydration state than in other forms of dehydration (see later). Therefore progressive dehydration associated with hypernatremia may be overlooked, even in babies with marked weight loss.[9]

Any form of dehydration is characterized by weight loss in the child. The magnitude of acute weight loss reflects the amount of water loss and is the best clinical indicator of the degree of dehydration. Weight losses in infants of 5%, 10%, and 15% (50, 100, and 150 /kg, respectively) classically correspond with mild, moderate, and severe degrees of dehydration. Older children and adults manifest symptoms at a somewhat lower degree of fluid loss than infants because the former have relatively smaller TBW and ECF volume, whereby weight losses of 3%, 6%, and 9% are better used as indicators of mild, moderate, and severe dehydration, respectively.[10]

In addition to weight loss, the reduction of ECF will cause a range of symptoms and physical signs, depending on the intensity and rate of the dehydration, such as decreased skin turgor, dry mucous membranes, sunken anterior fontanelle, cool and mottled skin with poor capillary refilling, oliguria, acceleration of heart rate and pulse, decrease in blood pressure and, in the most severe cases—most commonly, combined volume depletion with concomitant net sodium loss, as well—hypovolemic shock. A review of the literature to find out the precision and accuracy of symptoms, signs, and basic laboratory tests for evaluating dehydration in infants and children came to the conclusion that delayed capillary refill time, reduced skin turgor, and deep respirations, with or without an increase in rate, were the most useful clinical signs that predicted 5% hypovolemia; these parameters should be the basis of the initial assessment of dehydration in young children.[11]

In infants with hypernatremic dehydration, the reduction of the ICF volume may also cause fever and central nervous system (CNS) manifestations such as irritability, a high-pitched cry, seizures, and other neurobehavioral disturbances, reminiscent of febrile and neurologic illness. Hyperosmolality causes thirst and avidity for water. Elevations of hematocrit and plasma concentrations of total proteins, uric acid, and urea are biochemical indicators of dehydration, although they are not reliable parameters to assess its severity. Metabolic acidosis is usually present because of the frequent associated loss of bicarbonate and alkalinizing organic anions from the GI tract (non–anion gap [AG] acidosis) or anaerobic metabolism caused by poor peripheral perfusion (AG acidosis caused by lactate accumulation). A value of plasma bicarbonate below 17 mmol/L in the context of dehydration is suggestive of moderate or severe hypovolemia.[11]

Table 73.1 Pathogenesis of Hyperosmolar Disorders

Disorder	Features
Insufficient water intake	• Inability to obtain water: preambulatory infants, unconscious or disabled children, lack of water • Abnormalities of thirst—hypodipsia secondary to CNS disorders
Net loss of water in excess of solutes	• Renal losses—central and nephrogenic diabetes insipidus • GI losses: vomiting, diarrhea • Cutaneous losses: sweating, burns • Pulmonary losses: hyperventilation
High sodium intake + low intake of water free of solutes	Excessive administration of NaCl or $NaHCO_3^-$—parenteral nutrition, cardiopulmonary resuscitation, seawater intake

CNS, Central nervous system; *GI*, gastrointestinal.

The vast majority of cases of hypernatremic dehydration are caused by diarrheal illnesses, but two particular, albeit rare causes will be highlighted here, because they are specific to children and need to be considered in the differential diagnosis.

The Exclusively Breast-Fed Baby. Life-threatening hypernatremia with values up to 188 mmol/L has been reported in otherwise healthy newborn babies who were exclusively breast-fed.[12,13] These infants usually present between 10 and 21 days of age, and the spectrum of symptoms can range from minor nonspecific signs such as irritability, lethargy, and decreased urine output to jaundice, fever, anuria, and hypovolemic shock.[12] The cause is usually insufficient lactation and, even though many of these babies were regularly followed by health care professionals or even in the hospital, the extent of dehydration was typically missed until the dehydration had become severe. These babies are usually firstborns and inexperience with and sensitivities around breast-feeding, as well as early discharge from the hospital, may all be contributing factors.[13] Close attention to weight gain and feeding history during the first 2 to 3 weeks of life and support for first-time mothers may prevent or ameliorate this serious complication.

Nephrogenic Diabetes Insipidus. Nephrogenic diabetes insipidus (NDI) is a disease defined by renal resistance to arginine vasopressin (AVP) action, which results in an inability to concentrate urine adequately. Primary inherited NDI is discussed in detail in Chapter 44. Thus, this chapter will focus on the typical presentation and its fluid management.

Presentation. The osmotic load from breast milk is very low and, for this reason, urinary concentrating ability is usually not needed in the first weeks of life. This may explain the physiologic NDI that occurs in essentially all newborns, with full urinary concentrating ability being attained only by as late as 1 year of age.[5] Thus, most babies with NDI present later during the first year of life, or even later in childhood, when changes in diet present a higher osmotic load or when intercurrent illnesses lead to increased extrarenal water losses.[14] As discussed, apparent clinical signs of hypernatremic dehydration may not be noted until dehydration is severe. When infants affected with NDI present to medical attention with nonspecific symptoms, the families are often discharged without further investigation, because the medical staff is falsely reassured by the voluminous urine output typically provided in the history. Given that NDI is a rare disease, with most emergency physicians never encountering a single patient during their entire life of practice, and whereas dehydration from GI losses with a consequently reduced urine output in an otherwise healthy infant is a common condition, this mistake is understandable. Yet, it is important to realize that a large output of dilute urine in an infant with clinical signs of dehydration should be a severe warning sign, rather than reassuring, and plasma electrolyte levels should be assessed. In older children, reports of a constant craving for water, sometimes associated with unusual drinking behavior, such as from the bathtub, toilet, or flower vase, should alert the clinician to the potential of a urinary concentrating defect.

Fluid Management. Most patients with NDI maintain electrolyte homeostasis simply by drinking sufficient volumes of water to compensate for the renal losses. Problems arise when this is impaired, either because of an inability to drink sufficiently—for example, when the patient is vomiting or if oral intake is stopped, such as before a procedure requiring an anesthetic. In these cases, intravenous (IV) fluids need to be provided. Patients with NDI essentially lose pure water in the urine and thus should receive dextrose in water as IV fluid, just as they would otherwise drink water. Due to concerns over the development of hyponatremia in hospitalized children, most hospitals have discouraged or even banned hypotonic fluids for routine use.[15]

Although this makes sense for the child with a normal urinary concentrating ability, it poses a problem for the rare child with NDI. Administration of a fluid with a higher tonicity than the urine will increase the tonicity of the blood, leading to iatrogenic hypernatremia.[16] This is especially problematic if the child presents with hypernatremia. Because of concerns over rapid lowering of the plasma sodium level, with the consequent risk of brain edema and herniation, hypotonic fluids are usually avoided in hypernatremic dehydration.[17] Again, although this makes perfect sense in the child with a normal urinary concentrating ability, it can have disastrous consequences in the rare child with NDI because the administration of a fluid with a higher sodium concentration than the urine, such as 0.9% or even 0.45% saline, will worsen the hypernatremia.[18] Physicians caring for children with NDI need to be aware that the fluid needs of these children are different, and that they primarily need water, as highlighted by their own drinking habits. Concerns about the rapidity of restoring the water deficit should be accounted for by the rate of water replacement (see later), rather than by resorting to a fluid with a sodium and effective osmolality concentration that exceeds that of the child's urine. Increasing the osmotic load by provision of fluids high in sodium increases the obligate water losses needed for the kidneys to excrete this increased load.[14]

Secondary Forms. The most common form of NDI in adults is lithium-induced, but this is virtually nonexistent in children.[19] Instead, secondary NDI can be seen as a complication of other kidney diseases, such as Bartter syndrome, or in cystinosis.[20] Again, awareness by the clinician is key because treatment of these disorders usually involves supplementation with large doses of salt and/or the administration of salt-containing IV fluids. The unexpected development of hypernatremia should prompt investigations into a urinary concentrating defect and consequent adjustment of treatment.[21]

Hypernatremia From Excess Salt

The vast majority of cases of hypernatremia reflect dehydration—that is, a net water loss. However, there are rare cases of hypernatremia in children due to excess salt administration, which can occur inadvertently or with malicious intent, and recognizing these cases is important.[22] At risk are especially those children without free access to water, such as infants or tube-fed children, in whom the natural aversion to a high salt intake can be bypassed.[23] Because salt is an effective emetic and can cause osmotic diarrhea, affected children can present with GI symptoms, leading to a misdiagnosis of hypernatremic dehydration.[8] Most reported cases experienced recurrent episodes of hypernatremia, and there should be a high index of suspicion in such patients if the observed weight loss is not consistent with the expected change in weight if the hypernatremia was due to water loss.[8] Once suspected, the diagnosis is best established by calculation of the fractional excretion of sodium (FENa), and administered feeds and fluids should be ascertained.[9] Although the FENa level can also be elevated in hypernatremic dehydration, this will only be in the context of acute kidney injury.[13]

HYPOOSMOLALITY AND HYPONATREMIA: PATHOGENESIS AND CLASSIFICATION

Hypoosmolality indicates excess water relative to solute in the ECF. Because water moves freely between the ECF and ICF, this also indicates an excess of TBW relative to total body solute. Hyponatremia (plasma sodium concentration <135 mmol/L) and hypoosmolality (plasma osmolality <280 mOsm/kg H_2O) are typically used synonymously. However, high or normal osmolality with a low sodium concentration may be found when effective solutes other than sodium are present in the plasma, enlarging the osmolal gap. The latter is the difference between the measured osmolality and the calculated osmolality (see earlier); normal values in children are between 0 and 10 mOsm/kg H_2O.[24] Pseudohyponatremia, hyponatremia with normal osmolality, can be produced by marked elevation of plasma lipid or protein levels when measured by an indirect ion-selective electrode. This method is especially popular in pediatric laboratories because it requires only a very small sample volume, which is then diluted with water. In the calculation of the original sodium concentration in the plasma from the measured diluted sample, it is assumed that 93% of the plasma sample is water. If this assumption is wrong because of a high protein or lipid content, the calculated sodium concentration will be erroneously low.[25] Although the concentration of sodium per liter of plasma water is unchanged, the concentration of sodium per liter of plasma is decreased because of the increased nonaqueous portion of the plasma occupied by lipids or protein. This does not occur when using techniques such as direct potentiometry, in which the water content of the sample does not affect the measurement of sodium concentration, but that require larger sample volumes and is thus not commonly used in pediatric laboratories.

Children are at increased risk for developing hyponatremic encephalopathy because of the relatively larger ratio of brain to intracranial volume compared with adults. The brain reaches adult size by 6 years of age, but the skull does not reach adult size until 16 years of age.[26] The average plasma sodium concentration in children with hyponatremic encephalopathy has been found to be 120 mmol/L; in adults, it is 111 mmol/L. Hypoxia is also a major risk factor for the development of hyponatremic encephalopathy because it impairs the ability of the brain to adapt to hyponatremia, and hyponatremia in turn leads to a decrease in cerebral blood flow and arterial oxygen content.[27]

Briefly, hyponatremia may result from a loss of sodium in excess of water or from a gain of water in excess of sodium. For the kidneys, volume homeostasis is the most important priority; to maintain this, normal plasma osmolality and sodium concentration can be compromised. Thus, primary sodium depletion is accompanied by some degree of secondary retention of water by the kidneys in response to the resulting intravascular hypovolemia and consequent stimulation of AVP secretion.[28] Conversely, primary water retention can lead to hypervolemia, which in turn causes sodium losses.[25] Thus, both mechanisms (sodium loss and water retention) can coexist in the same patient, leading to frequent confusion in the diagnosis (e.g., cerebral salt wasting versus the syndrome of inappropriate antidiuretic hormone [ADH] release [SIADH]). Nevertheless, identifying the primary mechanism (sodium loss vs. water gain) is critical to institute appropriate treatment, which in turn will stop the secondary compensatory attempts by the kidneys to maintain volume homeostasis.

For this reason, patients with hyponatremia are best classified according to the status of the ECF volume (Table 73.2). The presence of ECF volume contraction is consistent with a primary sodium loss, whereas euvolemia or hypervolemia suggest primary water excess. In children with primary sodium loss, the FENa can help distinguish between renal (FENa > 1%) or extrarenal losses (<1%). Yet, FENa may only be increased acutely because compensatory mechanisms such as aldosterone will reduce renal sodium losses at the expense of potassium. Consequently, the fractional excretion of chloride (FECl) may be more informative of renal salt losses because both sodium and potassium in the urine need to be accompanied by an anion. In one study, an FECl value

Table 73.2 Clinical Classification of Hyponatremic Disorders in Children[a]

Contracted	Normal	Expanded
Renal solute loss	Administration of hypotonic solutions	Congestive heart failure
• Diuretics (thiazides)	SIADH	Cirrhotic liver disease
• Osmotic diuresis—hyperglycemia	Nephrogenic syndrome of inappropriate antidiuresis	Nephrotic syndrome
• Salt-wasting nephropathies	Severe hypothyroidism	Advanced oliguric renal failure
• Mineralocorticoid deficiency	Glucocorticoid deficiency	Excessive water intake
• Cerebral salt wasting		
Nonrenal solute loss		
• Gastrointestinal—vomiting, diarrhea, aspiration, fistula, stoma, third space		
• Cutaneous—sweating, burns, cystic fibrosis		

[a]According to the status of extracellular fluid volume.
SIADH, Syndrome of inappropriate antidiuretic hormone secretion.

more than 0.5% was consistent with a renal salt-wasting disorder.[29] Obviously, normal values for fractional excretion only apply in the setting of a normal GFR.[25]

In children with primary water excess, urine osmolality can help distinguish between renal water retention (e.g., SIADH—urine osmolality >100 mOsm/kg) and water intoxication (urine osmolality <100 mOsm/kg).[25]

TREATMENT OF HYPONATREMIA

The treatment of hyponatremia follows simple principles[30]:

- Hyponatremia due to water overload should be treated by restricting water intake and/or enhancing water excretion.
- In the rare case of hyponatremia due to sodium deficiency, sodium supplementation is appropriate.
- In a chronic case or case of unknown duration, without acute symptoms, extreme care needs to be taken to correct the hyponatremia slowly to avoid complications, such as osmotic demyelination (central pontine myelinosis).
- Regardless of the cause, in acute symptomatic hyponatremia (e.g., seizures, confusion, coma), a rapid increase in sodium concentration to a level considered acceptable (usually 125 mmol/L), and where symptoms improve, can be achieved by administering 3% saline. A simple approach of using a 2-mL/kg body weight bolus (maximal, 100 mL), which can be repeated, as needed, has been suggested.[30,31]

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE SECRETION

SIADH is characterized by hyponatremia less than 135 mEq/L, low plasma osmolality (<275 mOsm/kg H_2O in typical descriptions), and urine osmolality that is not at a minimum value of less than 100 mOsm/kg H_2O. The diagnosis requires the absence of clinical or biochemical signs of hypovolemia, diuretic use, and other causes of impaired free water excretion by the kidneys, such as nephrotic syndrome, hypothyroidism; adrenal insufficiency; renal, cardiac, or hepatic failure. Characteristically the urinary sodium concentration is greater than 20 mmol/L, reflecting the attempts by the kidneys to maintain volume homeostasis, because the primary water retention leads to volume overload, which is corrected by sodium excretion. This increased urinary sodium concentration often leads to diagnostic uncertainty or a misdiagnosis of salt wasting, which is then treated with sodium supplementation, typically resulting in hypertension.[25,32]

It is important to emphasize that the vast majority of cases of SIADH in children is acute and transient, resolving with the passage of time or after the underlying condition improves or heals. Thus, pediatricians usually do not cope with chronic states of hyponatremia. If the SIADH resolves while the patient has hyponatremia, AVP secretion will be suppressed by the hypo-osmolality, and a water diuresis will ensue. The treatment of choice for patients with acute SIADH is water restriction, which, if adhered to, usually achieves normalization of the plasma sodium concentration within 2 or 3 days. In those with persisting hyponatremia, treatment with AVPR2 antagonists, a vaptan or urea has been reported to be successful.[32–34]

HYPONATREMIA IN HOSPITALIZED CHILDREN

The prescription of hypotonic solutions as maintenance intravenous (IV) fluids in hospitalized children, according to the classic recommendations of Holliday and Segar,[35] has been challenged by a growing number of studies over the past 2 decades, which have drawn attention to the risk for severe neurologic complications and even death in hospitalized children who develop hyponatremia.[36] The incidence of acute hospital-acquired hyponatremia has been reported to be as high as 10% in a case-control study performed in a pediatric tertiary hospital.[37] Likewise, 11% of infants with severe bronchiolitis had serum sodium levels less than 130 mEq/L at the time of admission to the intensive care unit.[38] The retrospective incidence of postoperative hyponatremia among 24,412 pediatric patients with generally minor illnesses subjected to routine surgical procedures has been reported to be 0.34%, with a mortality rate of those so affected of 8.4%.[39]

Hospital-acquired hyponatremia is usually attributed to the administration of hypotonic fluids in the presence of simultaneous impaired free water excretion resulting from elevated AVP secretion by hemodynamic (e.g., effective circulating volume depletion) and nonhemodynamic (e.g., malignancies, CNS disorders, pulmonary diseases, medications, nausea, pain, stress) stimuli.[40] Excessive fluid administration is also considered a common contributing factor.[41] Thus, hospital-acquired hyponatremia typically reflects SIADH. To reduce the risk for hyponatremia, the British National Patient Safety Agency has recommended increasing the tonicity of IV fluids administered as maintenance therapy in children from 0.18% NaCl, as formerly used, to 0.45% NaCl, and prescribing isotonic solutions to hospitalized children with disorders entailing a high risk for elevated AVP secretion and ensuing hyponatremia.[42] Concerns that using solutions with higher salt content would result in an increased incidence of hypernatremia instead have not been confirmed.[43] Yet, considering that hyponatremia in the context of elevated ADH levels is due to an excess of water, rather than a deficiency of salt, a more physiologic measure to decrease the risk for hyponatremia would be the reduction of fluids to two-thirds of the normal recommended volume, except in dehydrated children.[44]

NEPHROGENIC SYNDROME OF INAPPROPRIATE ANTIDIURESIS

Nephrogenic syndrome of inappropriate antidiuresis is caused by activating mutations in AVPR2, thus constituting a mirror image to NDI.[45] The spectrum of symptoms has been reported to vary within the same family, ranging from infrequent voiding to incidentally noted hyponatremia to recurrent admissions with hyponatremic seizures.[46,47] Hyponatremia can manifest as early as the neonatal period, but usually is more prominent, once urine concentrating ability develops fully during the first year of life.[48] Unsurprisingly, given the X chromosome location of *AVPR2*, males are typically more severely affected than females. Administration of oral urea as osmotic agent at a dose titrated up from 0.25 up to 2 g/kg/day, given in up to four divided doses, has been shown to be effective in these children.[49]

Basis of Fluid Therapy for Dehydration in Children

The three steps in treating dehydration are as follows: (1) repletion of deficit, which means previous losses of fluids, and that can be estimated by the patient's weight loss; (2) maintenance therapy, which means physiologic requirements

Table 73.3 Features of Intravenous Fluids Commonly Used in Pediatrics

Solution	Osmolality (mOsm/L)	Sodium (mEq/L)	Osmolality (Compared With Plasma)	Tonicity (With Reference to Cell Membrane)
NaCl, 0.9%	308	154	Isosmolar	Isotonic
NaCl, 0.45%	154	77	Hyposmolar	Hypotonic
Glucose, 5%	278	0	Isosmolar	Hypotonic
Glucose, 10%	555	0	Hyperosmolar	Hypotonic
NaCl, 0.9% with glucose 5%	586	150	Hyperosmolar	Isotonic
Lactated Ringer's solution[a]	273	130	Hyposmolar	Isotonic
Hartmann's solution	278	131	Isosmolar	Isotonic
Human albumin, 5%	260	140	Hyposmolar	Isotonic

[a]Lactated Ringer's solution also contains 4 mEq/L of potassium as well as 1.4 mmol/L of calcium and 28 mmol/L of lactate.

of fluid and electrolytes; and (3) sustained provision of continuing extraordinary losses. Replacement of the deficit was traditionally carried out slowly, over 24 to 48 hours, particularly in the presence of hyponatremia or hypernatremia. In acute processes, usually from a GI origin, and when most of the fluid lost by the child comes from the ECF, current recommendations are more in favor of a policy of rapid repletion of the isotonic depletion to restore the ECF volume quickly on the basis of the following benefits—improved GI perfusion that allows earlier oral feeding, improved renal perfusion, and a lower morbidity and mortality rate.[44] If the child has signs of severe hypovolemia (see earlier), repletion therapy may begin with a rapid IV infusion of 10–20 mL/kg of isotonic 0.9% saline (Table 73.3), which can be repeated as needed until adequate perfusion is restored under careful monitoring of the patient. A recent large multicenter trial of fluid resuscitation in patients with septic shock, however, has cautioned against the use of fluid boluses, which was associated with a significant increase in mortality compared with children receiving only maintenance fluids without an initial bolus.[50]

Maintenance fluids used to be calculated following the Holliday-Segar method,[35] which estimates physiologic losses of water scaled to the metabolic rate. Yet, given the increased incidence of hyponatremia in children receiving IV fluids in an isotonic solution (see Table 73.3), is now considered an appropriate initial option.[15] The rate of fluid administration is determined by maintenance need, estimated fluid deficit (usually replaced over 24–48 hours), and ongoing losses. Key is ongoing close monitoring of clinical state, fluid balance, weight, and biochemistries to allow appropriate adaptation of fluid rate and composition, if needed. Under normal circumstances, the kidneys will be able to maintain volume and electrolyte homeostasis with almost any fluid administered, but problems arise when kidney function is impaired or if the kidneys receive inappropriate signals (e.g., as in SIADH).

In hypernatremic dehydration, the gravest danger is posed by a rapid decrease in plasma sodium concentration, leading to cerebral edema. A rate of decrease in the sodium concentration by 0.5 mmol/L/h or less is usually considered safe; fluid composition and/or the rate may need to be adjusted to maintain a safe normalization of sodium concentration, emphasizing again the importance of close monitoring of the patient.[17]

Rehydration therapy with oral solutions, such as the World Health Organization solution, should be attempted as an alternative to IV therapy in children with mild and moderate dehydration and good oral tolerance.[44] Even in patients with initial vomiting, the oral administration of small and frequent volumes of the solution often succeeds in correcting the dehydration.

POTASSIUM DISORDERS

Unlike adults, infants and children need to maintain a positive balance of potassium to grow normally because cellular mass depends on potassium as the major intracellular cation. Total body potassium increases from approximately 8 mmol/cm body height at birth to greater than 14 mmol/cm body height by 18 years of age, with the rate of accumulation of body potassium per kilogram of weight being more rapid in infants than in children and adolescents.[51] In vitro and experimental studies have indicated that the capacity of the neonatal kidney to secrete potassium is less than in the adult. The immature kidneys are relatively insensitive to aldosterone, which, in the clinical setting, correlates with the higher plasma concentrations of aldosterone and potassium and the lower ratio of urinary sodium to potassium characteristic of newborns and infants.[51]

HYPOKALEMIA

Hypokalemia is usually defined as a serum potassium concentration less than 3.5 mEq/L. Acute changes in the serum potassium concentration usually reflect modifications in the distribution of potassium between the extracellular and intracellular compartments. By contrast, alterations in the external potassium balance give rise to more chronic changes of plasma potassium concentration and lead to situations of true potassium deficit. Clinical conditions leading to sustained hypokalemia are listed in Table 73.4.

Potassium is critical for many important cell functions, so hypokalemia and potassium depletion may result in clinical manifestations in several organs and tissues (Table 73.5). In general, the same degree of hypokalemia is better tolerated when it results from a chronic potassium loss than from an acute decline, but both conditions may coexist in the same

Table 73.4 Most Relevant Causes of Hypokalemia in Pediatric Patients

Acquired Causes	Inherited Causes
Acute redistribution of potassium to the intracellular compartment	Hypokalemia with normal blood pressure
Metabolic alkalosis	Bartter syndrome
Insulin administration	Gitelman syndrome
Hypokalemic periodic paralysis	Epilepsy, ataxia, sensorineural deafness, and tubulopathy (EAST syndrome)
Prolonged lack of intake	Fanconi syndrome
Increased renal loss	Hypokalemia with hypertension
Drugs—diuretics, antibiotics, aminoglycosides, penicillin, amphotericin B, capreomycin	Congenital adrenal hyperplasia
Metabolic acidosis, including diabetic ketoacidosis	Primary hyperaldosteronism
Increased gastrointestinal loss	Liddle syndrome
Vomiting (hypertrophic pyloric stenosis)	Apparent mineralocorticoid excess
Diarrhea	Hypokalemia with acidosis
	Renal tubular acidosis (types 1, 2, and 3)

Table 73.5 Primary Manifestations of Hypokalemia and Potassium Deficit in Pediatric Patients

Organ System	Manifestation
Neuromuscular	• Skeletal muscle weakness • Muscular paralysis • Paralytic ileum • Muscle ischemia and rhabdomyolysis

Heart	• Electrocardiographic alterations
	  • Depression of S-T segment
	  • Low amplitude of T wave
	  • Appearance of U wave
	• Arrhythmias

Nutrition and growth	• Mild glucose intolerance • Growth retardation
Kidney	• Hypokalemic nephropathy • Polyuria-polydipsia

patient. Arrhythmias and marked weakness or paralysis of respiratory muscles are life-threatening symptoms that require urgent therapy. Growth retardation can be seen in children with hypokalemic tubular disorders.[29,52,53] Association with growth hormone (GH) deficit has been reported in a few of these patients.[54] Yet, these patients typically have complex abnormalities, typically including alkalosis and often elevated prostaglandin levels. To what degree the hypokalemia is responsible for the impaired growth remains unresolved. In one report, response to GH was only seen after concurrent treatment with a prostaglandin synthesis inhibitor.[29] In another study, amelioration of the alkalosis was associated with improved growth.[55] Similarly, long-standing hypokalemia has been associated with so-called hypokalemic nephropathy, which includes the development of renal cysts, chronic interstitial nephritis and, in the very long term, progressive loss of renal function. However, whether it is truly the hypokalemia causing this or other factors associated with the underlying disorder, such as hypovolemia or nephrocalcinosis, remains to be clarified. In one study, no association between the degree of hypokalemia and chronic kidney disease was seen.[56] Polyuria resistant to the administration of AVP and polydipsia (secondary NDI) has also been associated with hypokalemic disorders.[20,57] Yet, patients with Gitelman syndrome typically have preserved urinary concentrating ability, whereas patients with Bartter syndrome may have hyposthenuria, despite comparable potassium levels.[29] Moreover, of all forms of Bartter syndrome, type 3 (due to mutations in CLCNKB; see later) typically have the lowest plasma potassium levels, yet the best preserved urine-concentrating ability.[29,58] This suggests, that other factors such as hypercalciuria may actually be more relevant for the impaired urinary concentration.[21]

The cause of hypokalemia is usually apparent from the clinical history and global assessment of the patient. Excessive renal wasting of potassium can be identified by measuring potassium loss in urine collected for 24 hours or by calculating urinary indices in spot samples. Thus, in a hypokalemic child, fractional excretion of potassium (K):

$$\text{Fractional excretion of potassium} = (\text{urine K}/\text{plasma K}) \times (\text{plasma creatinine} \times 100/\text{urine creatinine})$$

above 30%, or a transtubular potassium gradient (TTKG):

$$\text{TTKG} = (\text{urine K}/\text{plasma K}) \times (\text{plasma osmolality} \times 100/\text{urine osmolality}))$$

greater than 6 is consistent with renal potassium wasting. In the clinical setting, the coexistence of hypokalemia of renal origin with alkalosis indicates enhanced sodium reabsorption in the collecting duct. If accompanied by arterial hypertension, this suggests primary salt retention, either due to stimulation of the mineralocorticoid receptor (e.g., renal vascular hypertension, salt-retaining forms of congenital adrenal hyperplasia, apparent mineralocorticoid excess) or from gain-of-function mutations in the mineralocorticoid receptor (pregnancy-associated hypertension) or the epithelial sodium channel ENaC (Liddle syndrome). Conversely, if the hypokalemic alkalosis is associated with normal or low blood pressure, this suggests a salt-wasting disorder (e.g., Bartter or Gitelman syndrome or extrarenal salt losses), with secondary activation of salt reabsorption in the collecting duct. A list of inherited disorders associated with hypokalemic alkalosis is provided in Table 73.4.

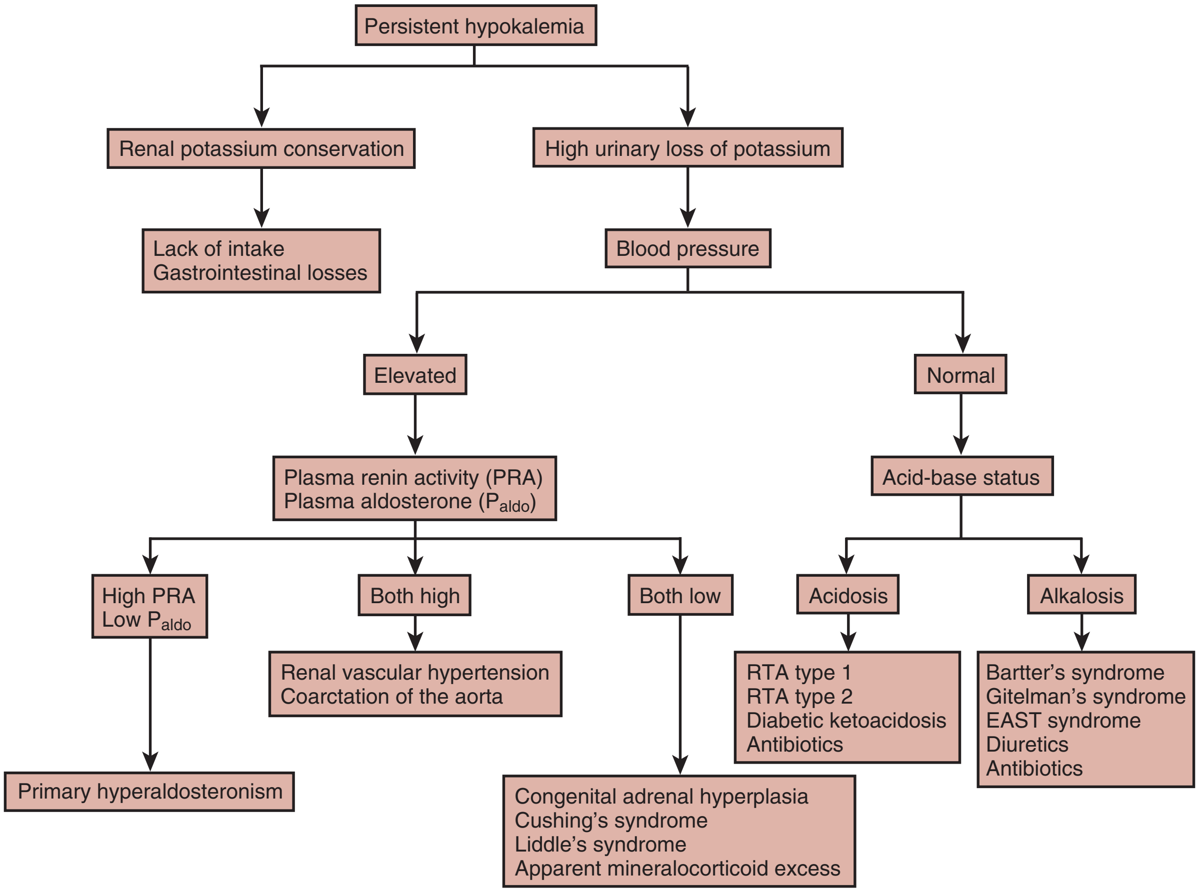

Fig. 73.1 Diagnostic approach to persistent hypokalemia in children. Transient hypokalemias are usually secondary to transcellular shifts of potassium ion. *EAST,* Epilepsy, ataxia, sensorineural deafness, and tubulopathy; *RTA,* renal tubular acidosis.

In contrast, if hypokalemia is associated with acidosis, this suggests renal tubular acidosis (RTA; proximal or distal, inherited or acquired; Fig. 73.1).

Hypokalemia secondary to internal potassium redistribution is transient and normalizes after the underlying disorder is corrected. Potassium deficit is treated with a potassium-rich diet and, if needed, oral potassium supplements, usually potassium chloride; potassium citrate is preferred in patients with concomitant acidosis. Patients with arrhythmias, respiratory paralysis, or rhabdomyolysis or those unable to take oral medications must be treated with IV potassium chloride at a maximum rate of 0.5 mEq/hg/hr and maximum concentration of 40 mEq/L in saline solutions. A higher concentration and rate of administration can be used in life-threatening situations under intensive care control and continuous electrocardiographic monitoring.[59]

INHERITED DISORDERS ASSOCIATED WITH HYPOKALEMIA

All disorders mentioned here are discussed in detail in the other relevant chapters; thus, mainly aspects specific to pediatric patients are considered here. We can distinguish three different forms of hypokalemia, based on the underlying pathophysiology, which leads to the following associated other homeostatic abnormalities (see Table 73.4): (1) alkalosis without hypertension; (2) alkalosis with hypertension; and (3) acidosis.

Hypokalemic Alkalosis Without Hypertension

Bartter Syndrome

Antenatal Presentation. The clinical manifestations of Bartter syndrome can already start during intrauterine life or immediately after birth; these include polyhydramnios, premature birth, polyuria, and loss of salt. A key aspect of Bartter syndrome is the impaired function of the juxtaglomerular apparatus (JGA), the site of tubular glomerular feedback (TGF), which acts as a "volume sensor" of the kidney.[60] Salt or, more precisely, chloride reabsorption in the macula densa, is a key signaling pathway for TGF, but is essentially short-circuited in Bartter syndrome due to the genetic knockout of salt reabsorption in the thick ascending limb of Henle's loop (TAL), which includes the macula densa.[55,61] The impaired chloride reabsorption leads to primary activation of cyclooxygenase-2 (COX-2), with the consequent formation of prostaglandin E2, hyperreninism, and afferent arteriolar dilation.[62,63] This aspect of Bartter syndrome is most prominent in types 1, 2, and 4 and is typically associated with antenatal presentation (Fig. 73.2).

It is presumed that in type 3, CLCNKA, which is also expressed in TAL, can compensate to some degree for the loss of CLCNKB function, as illustrated by the fact that combined mutations in both chloride channels are associated with a severe antenatal presentation (Bartter syndrome type 4b).[64] Yet, there is a wide spectrum of severity in all forms of Bartter syndrome and some patients with types 1, 2, or 4

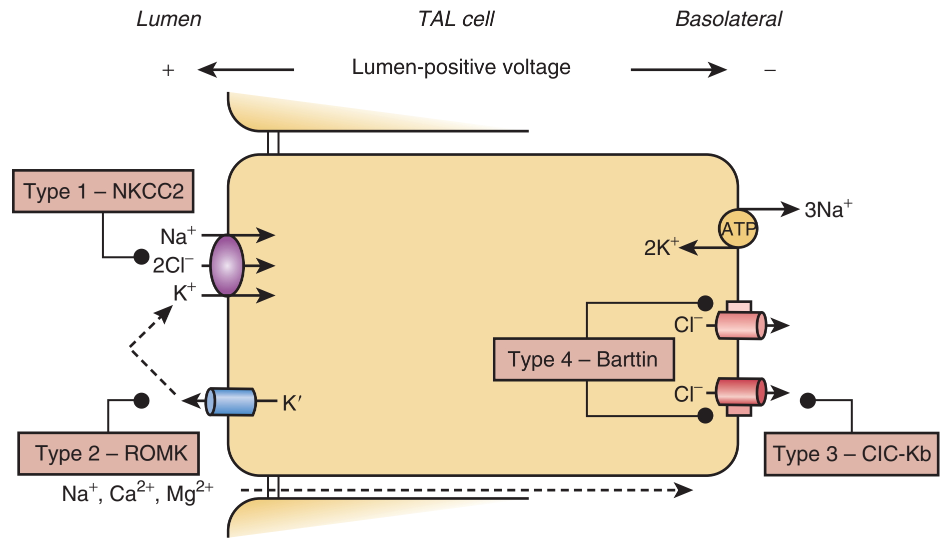

Fig. 73.2 Schematic representation of a cell of the thick ascending limb of the loop of Henle *(TAL)* showing the transport systems disturbed in the four types of Bartter syndrome. The bumetanide-sensitive Na^+-K^+-$2Cl^-$ cotransporter *(NKCC2)* and the renal outer medullary K^+ channel *(ROMK)* are depicted on the luminal membrane of the cell. The Na^+-K^+-ATPase and the chloride channels ClC-Ka and ClC-Kb with their Barttin subunit are represented on the basolateral side. See text for description of types 5 and 6 Bartter syndrome. *ATP,* Adenosine triphosphate.

can have mild forms and present later in childhood or even in adulthood, whereas up to one-third of patients with type 3 can have an antenatal or neonatal presentation.[62,65] Bartter syndrome is a salt-wasting disorder, so salt supplementation is a mainstay of treatment. In the antenatal form, this may include an IV infusion of saline solutions for prolonged periods of time to avoid dehydration and even death, but this will not be sufficient to suppress the prostaglandin production due to the dysfunction of TGF. This may explain why most patients with the antenatal presentation benefit clinically from the use of cyclooxygenase inhibitors, which ameliorate the polyuria and electrolyte wasting, as already noted by Bartter.[66] Due to the potential of severe complications of indomethacin, including bowel perforation and necrotizing enterocolitis, the use of the more selective COX-2 inhibitors, such as rofecoxib and celecoxib, has been proposed.[67,68] The association of these drugs with an increased incidence of cardiovascular complications in adults taking these medications for chronic pain has, however, dampened the enthusiasm, although it is unclear to what degree this risk applies to patients with Bartter syndrome and whether it outweighs the risks of indomethacin.[62,69]

A special aspect of Bartter syndrome type 2 is an initial presentation with hyperkalemia, sometimes resulting in an erroneous initial diagnosis of pseudohypoaldosteronism type 1 (see "Hyperkalemia," later).[70] This paradoxic feature can be explained by the expression of the renal outer medullary K^+ channel (ROMK), also in the collecting duct, where it mediates potassium secretion. The hyperkalemia resolves typically within the first week of life, presumably due to tubular maturation with the expression of other potassium channels, which can compensate for ROMK dysfunction in the collecting duct.[51]

An interesting aspect of antenatal regulation of tubular salt transport was revealed by the identification of mutations in MAGED2 as the cause of a severe, but transient, antenatal form of Bartter syndrome, with symptoms resolving completely within the first 3 to 4 weeks of life.[71] These patients showed evidence for reduced expression of both SLC12A1 and SLC12A3, suggesting that MAGED2 is involved in the regulation of expression of these transporters in utero.[72]

Some patients, typically with Bartter syndrome types 1 or 2, can have marked hyposthenuria (secondary NDI) and present a treatment dilemma—salt supplementation helps compensate for the renal salt losses, but excess salt in NDI worsens the urinary water losses.[20] Indeed, persistent hypernatremia, leading to the cessation of salt supplementation, has been reported in such patients.[29]

Classic Presentation. The patients included in Bartter's original report all presented in childhood.[73] For this reason, Bartter syndrome with a postnatal presentation is typically referred to as "classic" Bartter syndrome. It is usually associated with loss-of-function mutations in CLCNKB (type 3), but can be seen with all types, potentially due to residual function of the mutated proteins in some cases.[29,65,74,75] Despite the suggestion of a milder phenotype by the later presentation, patients with type 2 Bartter syndrome typically have the most severe electrolyte abnormalities and alkalosis. Supplementation of potassium salts rarely achieves normalization of plasma levels. Treatment with potassium-sparing diuretics can improve this, but obviously worsens the salt wasting and thus risks life-threatening hypovolemia.[55] Use of these drugs should thus be very carefully considered, especially in babies, who cannot self-regulate their salt and fluid intake, and is probably best reserved only for extreme cases with apparent complications from the electrolyte abnormalities, such as arrhythmia or paralysis.[62]

There are several other abnormalities associated with Bartter syndrome, including growth hormone deficiency, hypophosphatemia with renal phosphate wasting, and hyperparathyroidism.[54,76,77] Potentially, these are linked to the impaired electrolyte–acid-base balance and/or the elevated prostaglandin levels. Improvement of growth after treatment with celecoxib and amelioration of phosphate wasting after improvement of alkalosis has been reported in isolated cases.[29,55]

Gitelman Syndrome. Gitelman syndrome is arguably the most common tubulopathy, with an estimated prevalence of 1 in 40,000.[78] Although often considered primarily a disease of teenagers and adults, it is a common tubulopathy diagnosis also in children, and manifestations have been noted even in premature infants.[79,80]

Gitelman syndrome is biochemically characterized by hypokalemic metabolic alkalosis in association with hypomagnesemia and hypocalciuria, reflecting the side effects seen with thiazide treatment, consistent with the common molecular basis. Thiazides inhibit the sodium chloride cotransporter NCC, encoded by *SLC12A3*, recessive mutations which are the cause of Gitelman syndrome (Fig. 73.3).[81] Yet, the phenotype can be mimicked by mutations in *CLCNKB* (Bartter syndrome type 3) or *HNF1B*.[79,82] Moreover, there is variability in the severity of the phenotype, with some patients only showing minor electrolyte abnormalities, including normal plasma magnesium concentrations.[29,78]

Many patients are diagnosed incidentally, when blood tests are obtained for an unrelated indication. Other patients, however, present with other symptoms, including tetany, paresthesias, muscle cramps, salt craving, generalized weakness, and fatigue.[83] Interestingly, the magnitude of the electrolyte abnormalities does not necessarily correlate with the severity of symptoms as perceived by the patient. Some patients have only mild biochemical abnormalities but experience the disease as a disabling illness that disrupts their ability to have a normal life, whereas others have marked hypokalemic alkalosis and hypomagnesemia but feel well and are only occasionally troubled by symptoms, such as cramps.[84] Symptom severity, as reported by the patient, appears to be milder in children than in adults; in one pediatric study, 21% of children considered their symptoms as a moderate to big problem, compared with 45% of adults.[83,85]

There have been isolated reports of GH deficiency in Gitelman syndrome, but typically in patients without genetic confirmation of the diagnosis. It is thus unclear whether these patients had mutations in *SLC12A3* or in *CLCNKB* and thus a diagnosis of Bartter syndrome type 3, where GH deficiency has been reported recurrently.[86,87] Yet, also in molecularly confirmed series of patients, up to one-third of patients had a height less than 2 standard deviations (SD) below the norm.[85,87] Often, this is associated with pubertal delay and thus may not reflect the patient's final height.[87] As in Bartter syndrome, this may be related to the biochemical abnormalities. In one study, those patients deemed compliant with their treatment had significant improvement in growth compared with those found to be noncompliant.[85]

Gitelman syndrome is a disorder of salt reabsorption in the distal convolute. Salt transport in the TAL and thus the macula densa is intact. Therefore, in contrast to Bartter syndrome, TGF is intact, and hyperreninism should be suppressible by salt and volume expansion.[62] Patients typically have a pronounced salt craving, yet additional salt supplementation may still be beneficial.[78] Potassium chloride and magnesium supplements are often prescribed as well, depending on electrolyte abnormalities. As in Bartter syndrome, normalization of these abnormalities is frequently not achieved with oral supplementation, because increased plasma levels increase the filtered load and thus, typically, renal losses. Although severe complications of extreme electrolyte abnormalities have been reported, a stable subnormal concentration may still be safer than marked swings

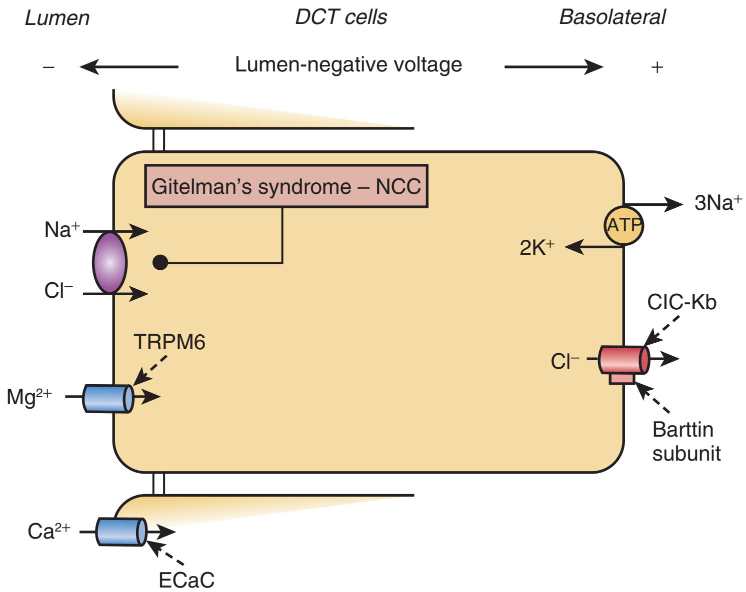

Fig. 73.3 Schematic representation of a cell of the distal convoluted tubule *(DCT)* showing the transport systems involved in the reabsorption of NaCl. The thiazide-sensitive Na^+-Cl^- cotransporter *(NCC)*, the magnesium-transporting channel TRPM6, and the epithelial calcium channel *(ECaC)* are depicted on the luminal membrane of the cell. The Na^+-K^+-ATPase and the chloride channel ClC-Kb with their Barttin subunit are represented on the basolateral side. NCC, ClC-Kb, and TRPM6 are expressed in the early downstream segment of the DCT, and ECaC is expressed in the late downstream portion of DCT. *TRPM6,* Transient receptor potential channel melastatin 6.

in plasma concentrations associated with large intermittent supplement doses.[78] For this reason, supplements should always be distributed in several doses throughout the day. Dissolving the salt supplements in a bottle of water, from which the patient drinks throughout the day, may help achieve more stable levels by providing frequent small doses. Moreover, a large dose can have side effects itself, such as diarrhea, vomiting (thus worsening electrolyte losses), and gastric ulcers. In an expert consensus statement, an "individual balance between improvement in blood values and side effects" was recommended, which may include target values well below the norm.[78]

EAST Syndrome. The EAST syndrome (*e*pilepsy, *a*taxia, *s*ensorineural deafness, and *t*ubulopathy), also known as SeSAME syndrome (*se*izures, *s*ensorineural deafness, *a*taxia, *m*ental retardation, and *e*lectrolyte imbalance) is rare, with approximately 30 patients described so far.[88–90] Mutations in the *KCNJ10* gene encoding a potassium channel expressed in the brain, inner ear, and basolateral membrane of the distal convoluted tubule in the kidney have been found in these children. Loss of function of this channel impairs salt reabsorption in the distal convolute and the renal manifestations thus mimic Gitelman syndrome, with hypokalemia, hypomagnesemia, and hypocalciuria.[88] Patients typically present in the first year of life with seizures, and blood tests at the time reveal the typical biochemical picture. Patients often receive enormous doses of potassium and magnesium supplementation out of concern that the electrolyte abnormalities are the cause of the seizures. Yet, experience so far suggests that seizures are independent of the biochemical alterations and rather reflect the important role of *KCNJ10* for potassium balance in the brain.[91] Indeed, whereas the biochemical abnormalities may worsen during school age, seizure activity spontaneously improves in most patients at this time.[92,93]

Claudin 10. Recessive mutations in *CLDN10*, part of the paracellular junction in the TAL, have been identified in two patients with a hypokalemic alkalotic salt-wasting phenotype, consistent with data from Cldn10 knockout mice that have shown a reduced sodium permeability in TAL.[94,95] Mutations in this gene may thus explain some patients with a Bartter-like tubulopathy yet no identified mutations in known genes.

Hypokalemic Alkalosis With Hypertension

In contrast to the salt-wasting disorders described previously, in which hypokalemic alkalosis reflects the secondary activation of the renin-aldosterone axis, the identical biochemical picture can also be seen in disorders with primary enhanced salt reabsorption in the collecting duct. These disorders are thus salt retaining and are consequently associated with hypertension (see Table 73.4). All are associated with enhanced salt reabsorption through the epithelial sodium channel ENaC in the principal cells of the collecting duct, either directly (Liddle syndrome) or through enhanced stimulation of the mineralocorticoid receptor (MCR). Most of these disorders are inherited in a dominant fashion and often are diagnosed later in childhood or even in adulthood. The exemption is apparent mineralocorticoid excess (AME), which is caused by recessive mutations in the enzyme HSD11B, which metabolizes cortisol to cortisone and thus normally protects the MCR from activation by cortisol.[96] The enzyme is also important in the placenta, and affected patients are typically born small for gestational age.[97] Further abnormalities include hypercalciuria and hyposthenuria, and patients may initially be misdiagnosed as having Bartter syndrome or NDI.[21] Hypertension is typically severe and left untreated, patients may die from cardiovascular complications, such as stroke or heart failure.[98]

Normalization of blood pressure and electrolyte abnormalities in all these salt-retaining disorders of the collecting duct can be achieved by pharmacologic blockade of ENaC (e.g., with amiloride or triamterene) and/or the MCR (e.g., spironolactone or eplerenone). Interestingly, recovery of urine-concentrating ability with such normalization has been reported in AME.[20]

Disorders Associated With Hypokalemia and Acidosis. Hypokalemia with acidosis is the typical biochemical constellation seen in either proximal (type 2) or distal (type 1) RTA. It will be discussed later.

HYPERKALEMIA

The same principles for the pathophysiology and assessment of hyperkalemia discussed in Chapter 17 apply in children, yet there are also some differences. Plasma potassium concentrations are higher in babies and young children, perhaps reflecting the increased potassium requirement for growth at this age. Consequently, the upper limit of normal is usually defined as a plasma potassium concentration of 6.5 mmol/L in newborns, which then gradually reduces to 5.5 mEq/L in older children.[99] Pseudohyperkalemia refers to those conditions in which the elevation in the measured serum potassium concentration is caused by potassium movement out of the cells during or after the blood specimen has been drawn, typically due to hemolysis caused by mechanical trauma during sample extraction. Considering the difficulties of obtaining a free-flowing blood sample, especially in premature babies, this is a complication encountered much more frequently in children than in adults. Causes of hyperkalemia are listed in Table 73.6.

The acute treatment of symptomatic hyperkalemia is discussed in Chapter 17. Typical treatment measures and doses are listed in Table 73.7. Aside from these emergency measures, treatment should focus on the underlying abnormality.

DISORDERS ASSOCIATED WITH HYPERKALEMIA

Pseudohypoaldosteronism

Pseudohypoaldosteronism (PHA) is a clinical syndrome caused by end-organ resistance to the effects of aldosterone, resulting in hyperkalemia, metabolic acidosis, and normal to high serum aldosterone levels. Due to the interdependence of sodium reabsorption and potassium and hydrogen ion secretion in the collecting duct, it is sometimes also referred to as type 4 RTA.[100] PHA may be primary (hereditary) or secondary (acquired).

Primary forms are subclassified into two distinct congenital syndromes, PHA type 1 (PHA1) and PHA type 2 (Gordon syndrome, or familial hyperkalemic hypertension; PHA2).

Table 73.6 Most Relevant Causes of Hyperkalemia in Pediatric Patients

Disorder	Relevant Causes
Pseudohyperkalemia	Improper collection of blood • Hematologic disorders—leukocytosis, thrombocytosis, spherocytosis
Transcellular shift of potassium	Acidosis • Insulin deficiency • Hyperosmolality • Exercise with nonselective beta-blockers • Familial hyperkalemic periodic paralysis
Increased potassium load	• From exogenous origin—pharmacologic supplements • From endogenous origin (cellular lysis)—burns, trauma, intravascular hemolysis, rhabdomyolysis, tumor mass destruction
Decreased urinary excretion	Renal failure Mineralocorticoid deficiency • Addison disease • Hypoaldosteronism Mineralocorticoid resistance • Types 1 and 2 pseudohypoaldosteronism Hyperkalemic drugs—potassium-sparing diuretics, trimethoprim, calcineurin inhibitors, renin-angiotensin-aldosterone system blockers

Importantly, PHA1 is a salt-wasting disorder and thus is associated with low-normal blood pressure or even hypovolemic shock. In contrast, PHA2 is a salt-retaining disorder associated with hypertension. Both disorders are discussed in detail in Chapters 17 and 44, so the focus here will be mainly on pediatric aspects of these disorders.

Secondary forms are typically associated with urinary obstruction, often in the context of a congenital urinary tract malformation and/or severe urinary tract infection.[101] Clinically, they resemble PHA1 and are typically seen in the first year of life. The electrolyte and acid-base abnormalities resolve with treatment of the obstruction or infection.

Pseudohypoaldosteronism Type 1. PHA1 may be renal or systemic. The renal form occurs more frequently and is caused by heterozygous inactivating mutations in the *NR3C2* gene encoding the human MCR.[102] Patients typically present in the first weeks of life, with failure to thrive and frequent episodes of vomiting and dehydration. Biochemistries reveal hyponatremia, hyperkalemia, and metabolic acidosis, although there is considerable variability in severity. Treatment consists of sodium supplementation (in the form of sodium chloride and sodium bicarbonate) in amounts sufficient to normalize the biochemistries and maintain normovolemia. Plasma electrolytes typically normalize, even without supplementation, as early as 6 months of age, although elevated concentrations of aldosterone persist.[79,103]

Systemic PHA1 follows an autosomal recessive transmission and is caused by mutations in the *SCNN1A*, *SCNN1B*, or *SCNN1G* genes encoding the α-, β-, and γ- subunits of ENaC, respectively.[104] ENaC is expressed in the epithelial cells of the renal collecting ducts, respiratory airways, colon, and salivary and sweat glands; in addition to the severe renal salt-wasting, with life-threatening hypovolemia, extreme hyperkalemia, and metabolic acidosis, patients can suffer from a cystic fibrosis-like lung involvement, as well as a miliaria rubra-like rash.[105]

Affected children typically present in the first few days of life, with failure to gain weight and inadequate feeding, and are often severely hypovolemic or even in shock, with dramatic hyperkalemia, sometimes exceeding 10 mmol/L.[105] Acute treatment focuses on volume restoration and normalization of the acidosis, with an IV infusion of saline and sodium bicarbonate. Potassium levels improve with expansion of the intravascular volume and correction of the acidosis, but in the long term, sodium exchange resins can be very helpful to provide a steady source of sodium supplementation and potassium removal. Sodium supplementation with doses up to 50 mmol/kg/day have been reported but, despite this, patients can experience acute episodes of hypovolemia, typically associated with intercurrent infections.[106] In our own clinical practice (unpublished), we typically use gastrostomies to facilitate the regular administration of supplements and feeds and sometimes use port-a-caths to ensure emergency IV access during a hypovolemic crisis. Management typically becomes easier as the child reaches school age, with episodes of hypovolemic crisis becoming rare and lung manifestations improving. Nevertheless, lifelong salt supplementation is needed.[105]

Pseudohypoaldosteronism Type 2. PHA2 (Gordon syndrome, or familial hyperkalemic hypertension) is characterized by hyperkalemia and hyperchloremic metabolic acidosis, with normal to high circulating levels of mineralocorticoids but suppressed levels and activity of renin. It is distinguished clinically from PHA1 by its distinct lack of sodium wasting and volume depletion. Instead, salt retention and hypertension are the hallmarks of this syndrome.[107] So far, four genes *(WNK1, WNK4, KLHL3,* and *CUL3)* have been identified as causative, all inherited in an autosomal dominant fashion. In addition, mutations in *KLHL3* can also cause a recessive form of the disorder.[108] Mutations in all four genes directly or indirectly stimulate activity of NCC (SLC12A3) the sodium-chloride cotransporter in the distal convoluted tubule (DCT).[109] In this way, PHA2 pathophysiologically and clinically constitutes the mirror image to Gitelman syndrome, caused by loss of function of NCC.

The phenotypic variability is high. Several patients are identified incidentally because of hyperkalemia noted on an unrelated blood test or because of family screening. Hypercalciuria is also frequently noted. Hypertension is present in approximately two-thirds of patients at the time of diagnosis, is often absent early in life, and is believed to develop eventually in all patients with time.[110] Less consistent features are age of apparent symptoms (neonate to adulthood), and there appears to be some genotype-phenotype correlation, with mutations in *CUL3* typically causing the most severe form, with earliest diagnosis and most severe electrolyte abnormalities.[111] In addition, patients with mutations in *CUL3* typically have

Table 73.7 Management of Severe Hyperkalemia in Children[a]

Parameter	Route of Administration	Dose	Onset of Action	Comments
Stabilization of Cardiac Membranes				
Calcium gluconate, 10% solution	IV	0.5–1 mL/kg over 10 min	Immediate	Repeat one dose after 5 min if ECG changes persist; cardiac monitor
Movement of Potassium Into Cells				
Glucose + Regular insulin	IV	10% dextrose at 5 mL/kg/hr + insulin, 0.1–0.2 U/kg/hr	15–30 min	Monitor blood glucose level at least hourly.
Salbutamol	Nebulized	0.1–0.3 mg/kg	20–30 min	Doses may be repeated as often as needed; effect lasts for up to 2 hr; tachycardia is main side effect.
	IV	4 µg/kg over 10–20 min	5–10 min	
Sodium bicarbonate	IV	2–3 mEq/kg over 30 min	15–30 min	Do not give in the same IV line as calcium; minimum effect unless the patient has acidosis.
Removal of Potassium From the Body				
Cation exchange resins	PO	1 mo–8 yr: 125–250 mg/kg (max, 15 g)	1–2 hr	3-4 times daily
	Rectal	Neonate to 18 yr: 125–250 mg/kg	1–2 hr	Repeat doses every 6 hr if needed; colonic irrigation to remove resin after 6-12 hr. Dilute 1 g resin with 5–10 mL methylcellulose or water. Do not give within 1 wk of surgery. Do not use enema with sorbitol, and keep in mind that sodium polystyrene sulfonate use, both with and without sorbitol, may be associated with fatal gastrointestinal injury.
Furosemide	IV	1–2 mg/kg	5 min	Repeat dose every 2 hr if needed; monitor fluid balance.
Hemodialysis or peritoneal dialysis				Hemodialysis is more effective.

[a]The measures listed here can be carried out simultaneously. The choice of therapy to be used depends on the severity of the symptoms and the particular clinical situation.
ECG, Electrocardiographic, *IV,* intravenous; *PO,* orally.

failure to thrive or growth impairment, and mutations are often de novo, suggesting an impaired reproductive fitness.[108] Although PHA2 is rare, diagnosis is important, not only for genetic counseling, but also to initiate appropriate treatment. A key feature of PHA2 is the complete reversal of biochemical abnormalities and hypertension by treatment with a thiazide, which, in fact, can be used as a diagnostic feature. There is also an acquired form of PHA2, associated with tacrolimus use.[112] Again, recognition of the combination of hyperkalemia and hypertension is important because patients with the acquired form also respond to thiazide treatment.

HYDROGEN ION DISORDERS

ACID-BASE EQUILIBRIUM IN CHILDREN

Acid-base homeostasis is maintained by the kidney by adjusting acid excretion to the acid load. The acid load is derived chiefly from the diet, specifically sulfur-containing amino acids, which are metabolized to sulfuric acid.[113] Net acid excretion (NAE) is defined as the sum of titratable acid and ammonium (NH_4^+) minus urine HCO_3^-. This emphasizes the importance of tubular reabsorption of filtered HCO_3^- to enable NAE and highlights the two key factors that can lead to acidosis—net loss of HCO_3^- or the net gain of acid. The AG (usually calculated as $Na^+ - (Cl^- + HCO_3^-) = AG$) can be used to distinguish between these two forms. Because HCO_3^- loss has to be accompanied by a cation, typically Na^+, the AG is unchanged. In contrast, with net acid gain (e.g., lactic acid), the excess acid is buffered by HCO_3^-, leading to an increased AG because Na^+ and Cl^- are unchanged, yet HCO_3^- is decreased. Thus, net HCO_3^- loss is associated with a normal AG acidosis, whereas net acid gain leads to an increased AG. A normal AG is typically 8 to 12 mmol/L, but depends mainly on the albumin concentration, the main anion missing from the calculation.[114] The AG falls approximately 2.3 mmol/L for every 10-g/L reduction in serum albumin concentration.[115] A list of causes of acidosis according to AG is given in Table 73.8.

Table 73.8 Metabolic Acidosis in Children According to the Serum Anion Gap

Serum Anion Gap	Features
Normal	Loss of bicarbonate in the stool • Diarrhea • Digestive fistulas or surgical drainage • Drugs—$CaCl_2$, $MgCl_2$, cation exchange resins Loss of bicarbonate in urine • Mild to moderate renal failure (greater tubulointerstitial than glomerular damage) • Renal tubular acidosis • Extracellular volume expansion

High	Advanced renal failure (uremic acidosis)
	• Ketoacidosis
	  ◦ Diabetes mellitus
	  ◦ Inborn errors of metabolism
	  ◦ Prolonged fasting
	• Lactic acidosis
	  ◦ Tissue hypoxia
	  ◦ Inborn errors of metabolism
	Toxins—salicylate, methanol, ethylene glycol

Table 73.9 Classification of Renal Tubular Acidosis in Children by Cause

Type	Features
Proximal renal tubular acidosis (RTA) (type 2)	Primary • Autosomal recessive: with ocular abnormalities Secondary • Component of renal Fanconi syndrome • Drugs and toxins (e.g., ifosfamide, carbonic anhydrase inhibitors, heavy metals)
Distal RTA (type 1)	Primary • Autosomal dominant • Autosomal recessive, with or without sensorineural deafness Secondary • Autoimmune diseases—Sjögren syndrome, rheumatoid arthritis, hyperglobulinemia • Drugs and toxics (e.g., amphotericin B, lithium salts, amiloride, toluene)
Mixed proximal and distal RTA (type 3 RTA)	Deficiency of carbonic anhydrase II (associated with osteopetrosis and mental retardation)
Hyperkalemic RTA (type 4)	Hypoaldosteronism • Pseudohypoaldosteronism • Drugs and toxics (e.g., potassium salts, heparin, spironolactone, triamterene, amiloride, trimethoprim, indomethacin, ACEi, ARB, cyclosporin A, tacrolimus)

ACEi, Angiotensin-converting enzyme inhibitor; *ARB,* angiotensin receptor blocker.

The NAE per kilogram of body weight is higher in infants than in adults and less in infants fed human milk than in infants fed formulas. This reflects the higher metabolism of infants associated with growth and the higher content of sulfur-containing amino acids in formula milk.[116]

Except for the few hours during and after birth, the blood pH during infancy and childhood remains relatively constant. Values are equivalent to normal adult values at 3 to 5 years of age.

Metabolic acidosis is diagnosed by a low plasma HCO_3^- concentration (<22 mmol/L in children and <20 mmol/L in infants) and a low blood Pco_2 (<40 mm Hg in children and <35 mm Hg in infants). If the pH is also low, this is termed *acidemia.*

RENAL TUBULAR ACIDOSIS

The term *renal tubular acidosis* refers to a group of clinical entities in which normal AG hyperchloremic metabolic acidosis occurs as a result of impaired proximal tubular reabsorption of HCO_3^- or distal secretion of H^+. Consequently, RTA can be classified as proximal (type 2) or distal (type 1; Table 73.9). In addition, there is a rare combined form (type 3) caused by recessive mutations in the gene encoding carbonic anhydrase II (see later). The so-called hyperkalemic RTA (type 4) is really a disorder of impaired salt reabsorption in the collecting duct, with consequent hyperkalemia and acidosis, and is discussed previously (see "Pseudohypoaldosteronism"). This section focuses on the primary congenital inherited forms of RTA because they are characteristic of the pediatric age; acquired forms are more prevalent in adults. As before, the focus will be primarily on aspects relevant to pediatrics; these inherited forms are discussed in more detail in Chapter 44.

PROXIMAL TYPE 2 RENAL TUBULAR ACIDOSIS

Proximal RTA results from decreased reabsorption of HCO_3^- in the proximal tubule (Fig. 73.4). Most cases in children are seen in the context of a generalized dysfunction of the proximal tubule (renal Fanconi syndrome). Thus, aside from RTA, these children typically display the whole spectrum of a proximal tubulopathy, such as renal wasting of phosphate, glucose, amino acids, and low-molecular weight proteins.[117]

RTA as the only manifestation of proximal tubular dysfunction is an exceedingly rare disorder and is caused by mutations in *SLC4A4* encoding the basolateral Na^+-HCO_3^- cotransporter (NBC1), which also plays a major role in HCO_3^- transport in the ocular lens epithelium.[118,119] Consequently, patients typically have eye abnormalities, such as band keratopathy, glaucoma, or cataract, in addition to the proximal RTA. Hence, the name "Proximal RTA with Ocular Abnormalities" (Online Mendelian Inheritance in Man [OMIM] no. 604278). Additional clinical features may include enamel defects of the permanent teeth, impaired intellectual capacity, calcification of the basal ganglia, hypothyroidism, and hyperamylasemia.[118] The mainstay of treatment of the acidosis is alkali supplementation yet, given the vast amount of physiologic HCO_3^- reabsorption in the proximal tubule, the normalization of plasma HCO_3^- concentration is typically impossible when

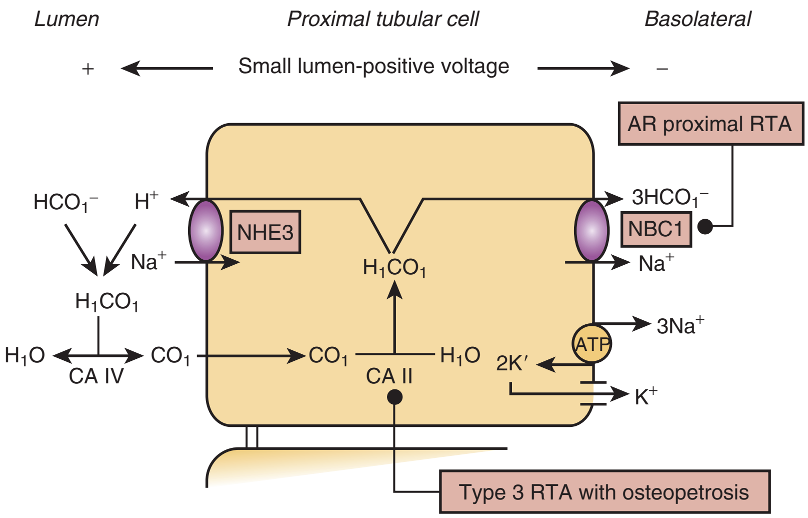

Fig. 73.4 Schematic representation of the reabsorption of bicarbonate in a cell of the proximal tubule showing the transport systems involved in the genesis of primary proximal renal tubular acidosis *(RTA)*. The Na^+-H^+ exchanger isoform 3 *(NHE3)* is depicted on the luminal membrane of the cell. The Na^+-K^+-ATPase and Na^+-HCO_3^- cotransporter *(NBC1)* are represented on the basolateral side. *AR,* Autosomal recessive; *ATP,* adenosine triphosphate; *ATPase,* adenosine triphosphatase; *CA II,* intracellular carbonic anhydrase; *CA IV,* luminal carbonic anhydrase.

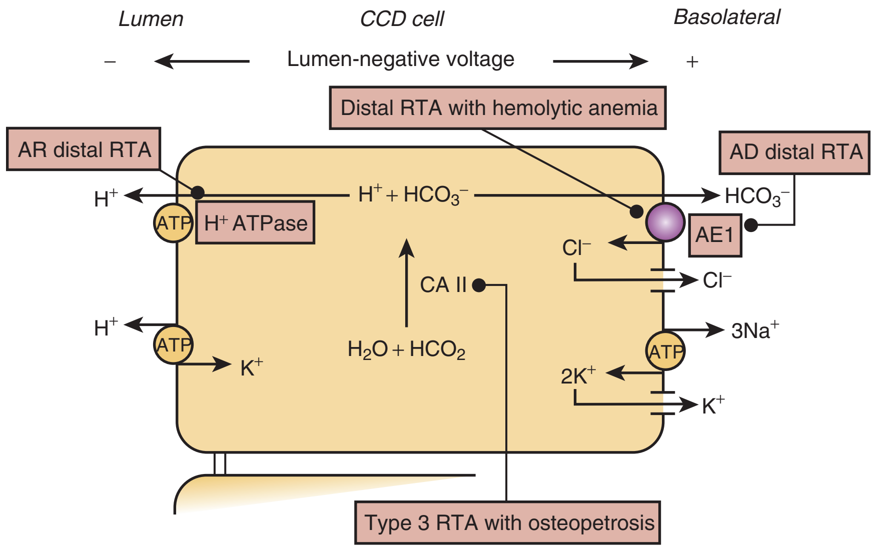

Fig. 73.5 Schematic representation of the system of acidification and regeneration of bicarbonate in an α-intercalated cell of the cortical collecting duct *(CCD)* showing the transport systems involved in the genesis of primary distal renal tubular acidosis *(RTA).* The H^+-ATPase pump and H^+-K^+-ATPase are depicted on the luminal membrane of the cell. The Na^+-K^+-ATPase and Cl^--HCO_3^--exchanger *(AE1)* are represented on the basolateral side. *AD,* Autosomal dominant; *AR,* autosomal recessive; *ATP,* adenosine triphosphate; *ATPase,* adenosine triphosphatase; *CA II,* intracellular carbonic anhydrase.

this is impaired. Patients with proximal RTA have a lower threshold for HCO_3^- reabsorption—that is, the plasma concentration at which bicarbonate is wasted into the urine is lower than normal. Consequently, by increasing the plasma concentration with alkali supplementation, renal HCO_3^- losses are augmented. Nevertheless, improvement of growth velocity with alkali supplementation has been reported.[118]

PRIMARY DISTAL TYPE 1 RENAL TUBULAR ACIDOSIS

Primary distal type 1 RTA is the most common RTA found in children. It results from defective urinary acidification in the cortical collecting tubule secondary to mutations in genes involved in the elimination of H^+ and associated regeneration of HCO_3^- (Fig. 73.5). Autosomal dominant distal RTA is caused by mutations in the *SLC4A1* gene, leading to loss of function by abnormal processing or trafficking of the Cl^--HCO_3^- exchanger (AE1) located at the basolateral surface of α-intercalated cells and in the erythrocytes.[120,121] Mutations in *SLC4A1* also cause the dominant red cell morphologic diseases of hereditary spherocytosis and ovalocytosis, although the association of these entities with distal RTA is rare.[121]

The apical proton pump H^+-ATPase that transfers H^+ into the urine (see Fig. 73.5) is composed of at least 13 different subunits.[122] Mutations in the gene *ATP6V1B1* encoding the B1 subunit of H^+-ATPase are responsible for a form of autosomal recessive distal RTA associated with sensorineural deafness.[123] H^+-ATPase containing the B1 subunit is also involved in maintaining a pH of 7.4 and low Na^+ and high

K^+ concentrations in the endolymph, explaining the concomitant deafness. Although usually congenital and severe, in some individuals hearing impairment is delayed in onset.[124] Recessive mutations in *ATP6V0A4* encoding the a4 subunit of H^+-ATPase are responsible for another form of autosomal recessive distal RTA.[125] It is initially associated with preserved hearing; however, long-term follow-up of patients with distal RTA caused by *ATP6V0A4* mutations has demonstrated that some of them develop hearing loss consistent with the expression of *ATP6V0A4* within the human cochlea.[126] Both *ATP6V1B1* and *ATP6V0A4* are regulated by FOXI1, a transcription factor important for the function of the inner ear, as well as the intercalated cells in the collecting duct.[127] Recessive mutations in *FOXI1* have also been associated with distal RTA (dRTA) and deafness.[128]

Clinical Presentation

Patients with recessive forms of dRTA typically present in the first year of life, with failure to thrive and a history of polydipsia/polyuria. Nephrocalcinosis is usually present and marked.[124,129] An associated proximal tubular dysfunction, suggesting an erroneous diagnosis of renal Fanconi syndrome, is common, but disappears with treatment of the acidosis. Patients typically have a urine-concentrating defect, sometimes attributed to the hypercalciuria and/or hypokalemia, although it persists even after normalization of the biochemical abnormalities and thus may reflect the nephrocalcinosis.[21]

In contrast, patients with the dominant form are increasingly diagnosed before the development of overt symptoms by genetic screening based on the family history. With adequate treatment and compliance, complications such as nephrocalcinosis and stones can be minimized and potentially completely avoided.[129]

Treatment

In contrast to most tubulopathies, where symptoms can only be ameliorated, treatment of dRTA is usually rewarding for patients and clinicians because oral alkali supplementation normalizes the biochemical abnormalities and induces catch-up growth. Citrate or bicarbonate supplements are typically used, ideally as a potassium salt. If given as a sodium salt, additional potassium supplementation is often needed. Large doses of up to at 9 mmol/kg/day of alkali equivalent may be needed, especially in infants at the beginning of treatment.[129] Frequent dosing, every 4 to 6 hours, is important to maintain normal values of the serum HCO_3^- concentration throughout the day. In infants, distribution of the daily dose throughout all feeds is often the easiest. Development of longer acting formulations allowing twice-daily administration should facilitate treatment in the future.[130]

HYPERKALEMIC TYPE 4 RENAL TUBULAR ACIDOSIS

Hyperkalemic (type 4) RTA is found together with aldosterone deficiency or aldosterone resistance and thus primarily reflects a disorder of sodium transport. It was discussed earlier (“Pseudohypoaldosteronism”).

MAGNESIUM DISORDERS

The regulation of serum magnesium (Mg) concentration depends primarily on the kidney. With normal magnesium intake, approximately 95% to 97% of filtered magnesium is reabsorbed along the nephron. In contrast to most other ions, the proximal tubule contributes the relatively smallest part to magnesium reabsorption, only about 5% to 15%, although it is of note that a much larger fraction of filtered magnesium is reabsorbed in the proximal tubule in newborns. About 50% to 70% is reabsorbed in the TAL, where it diffuses passively through the paracellular pathway. Finally, 5% to 10% of Mg is reabsorbed in the early DCT. This segment determines the final urinary magnesium concentration by active transcellular absorption via the TRPM6 channel.[131] The body of an adult contains approximately 24 g of magnesium, of which only a small fraction, around 60 mg, is present in the plasma, where it is available for measurement.[132]

HYPOMAGNESEMIA

In the clinical setting, low serum magnesium concentrations indicate magnesium deficiency, although values in the normal range do not rule out the possibility of a total body deficit, compensated for by the release of magnesium from the bone.[133] In the presence of hypomagnesemia, normal kidneys retain magnesium by decreasing its excretion to as little as 0.5% of the filtered load. In clinical practice, in the context of hypomagnesemia, a fractional excretion of magnesium (FEMg) of 2% is typically considered normal, whereas more than 4% is considered pathogenic.[134] For the calculation of FEMg, the serum concentration of magnesium is usually multiplied by 0.7 to use the filtered fraction only and exclude the estimated protein-bound fraction.[131] Of note, with markedly decreased serum levels, FEMg may be falsely normal if the remaining intact pathways are able to cope with the small filtered load. Assessment while the patient is receiving magnesium supplements is better suited to determine the presence or absence of renal magnesium wasting in uncertain cases.

Known genetic causes of hypomagnesemia are listed in Table 73.10 and depicted in Fig. 73.6. Based on clinical and pathophysiologic criteria, hypomagnesemia disorders can be classified into four groups[131]:

1. Hypercalciuric hypomagnesemias, affecting magnesium reabsorption primarily in the TAL
2. Gitelman-like hypomagnesemias, affecting transport in DCT in general
3. Mitochondrial hypomagnesemias, which may affect magnesium and other transport pathways along the entire nephron and consequently can have a very variable phenotype
4. Other hypomagnesemias, which include several forms of isolated or syndromic hypomagnesemia not fitting into the previous groups

CLINICAL MANIFESTATIONS

Hypomagnesemia is often asymptomatic, and clinical manifestations of magnesium deficiency are generally not manifested until the serum magnesium concentration decreases below 0.4 mmol/L, although the appearance of symptoms may depend on the rate of development of magnesium deficiency or on the total body deficit. Initial symptoms are often nonspecific, such as fatigue or cramps, which may not

Table 73.10 Main Causes of Hypomagnesemia in Children

Type of Hypomagnesemia	Description
Primary Inherited Disorders	
1. Hypercalciuric hypomagnesemias	Familial hypomagnesemia with hypercalciuria and nephrocalcinosis (*CLDN16,19*)
2. Gitelman-like hypomagnesemias	Gitelman syndrome (*SLC12A3*) Bartter syndrome (types 3 and 4, *CLCNKB, BSND*) EAST syndrome (*KCNJ10*) Autosomal dominant hypomagnesemia (*FXYD2*) HNF1B nephropathy Neonatal hyperphenylalaninemia and primapterinuria (*PCBD1*)
3. Mitochondrial hypomagnesemias	Mitochondrial metabolic syndrome (*tRNA*) Kearns-Sayre and Pearson syndrome (*SARS2*)
4. Other hypomagnesemias	Hypomagnesemia with secondary hypocalcemia (*TRPM6*) Hypomagnesemia seizures and mental impairment (*CNNM2*) Autosomal dominant hypomagnesemia (*KCNA1*) Hypomagnesemia with abnormalities in EGF signaling (*EGF, EGFR*) Hypomagnesemia with Kenney-Caffey syndrome (*FAM111A*)
Secondary Disorders	
Decreased gastrointestinal absorption	Malabsorptive syndromes • Vomiting and diarrhea
Increased urinary excretion	Drugs • Diuretics • Calcineurin antagonists • Others—cisplatinum, aminoglycosides, amphotericin B
Miscellaneous	"Hungry bone," low-birth-weight newborn, infant of diabetic mother

be presented to medical attention or not investigated. Often, patients may present with tetany. Magnesium depletion inhibits parathyroid hormone (PTH) secretion and causes resistance to PTH action, leading to hypocalcemia and consequent neuromuscular hyperexcitability and tetany or even convulsions.[135] Cardiac manifestations of hypomagnesemia can include prolongation of the QT interval and a wide range of arrhythmias.

HYPERCALCIURIC HYPOMAGNESEMIA: FAMILIAL HYPOMAGNESEMIA WITH HYPERCALCIURIA AND NEPHROCALCINOSIS

Familial hypomagnesemia with hypercalciuria and nephrocalcinosis (FHHNC) is an autosomal recessive primary tubulopathy caused by loss-of-function mutations in *CLDN16* or *CLDN19*.[136,137] These genes encode two proteins, claudin 16 (formerly paracellin 1) and claudin 19, which are expressed in the tight junction in TAL and are critical for magnesium and calcium permeability.[138,139] Claudin 19 is also important in the eye, explaining the additional ocular involvement, such as macular colobomata, significant myopia, and horizontal nystagmus.

Most common presenting manifestations of FHHNC are recurrent urinary tract infections and polyuria and polydipsia. Hematuria, abdominal pain, urolithiasis, sterile leukocyturia, growth failure, and seizures are also frequent.[140,141]

Chronic kidney disease is often present at diagnosis, and progression is not affected by magnesium supplementation.

GITELMAN-LIKE HYPOMAGNESEMIA

This group includes salt-wasting disorders of the DCT, such as Gitelman and EAST syndrome, as well as Bartter syndrome types 3 and 4.[131] Although not invariably present, hypomagnesemia is a typical feature in these disorders, otherwise characterized by hypokalemic hypochloremic alkalosis (see previously, "Hypokalemic Alkalosis Without Hypertension"). Dominant mutations in *FXYD2*, encoding the gamma subunit of the Na-K-ATPase in DCT, can cause a similar picture, albeit reported in only three families so far, putatively descended from a single founder.[142] *FXYD2* is regulated by the transcription factor HNF1B and, consequently, mutations in HNF1B can also cause a similar phenotype.[143] HNF1B, in turn, is regulated by *PCBD1*, which affects dimerization. *PCBD1* encodes an enzyme involved in the metabolism of aromatic amino acids, and loss-of-function mutations cause neonatal hyperphenylalaninemia and primapterinuria.[144] Reexamination of patients with such mutations has revealed the presence of maturity onset diabetes of the young (MODY) and hypomagnesemia, mimicking the HNF1B phenotype.[145]

MITOCHONDRIAL HYPOMAGNESEMIA

An association between mitochondrial dysfunction and hypomagnesemia was first highlighted during the investigation of a large pedigree affected with metabolic syndrome and hypomagnesemia, which cosegregated with a mutation in the mitochondrial tRNA(Ile) gene.[146] Hypomagnesemia has been further recognized in other mitochondrial diseases, specifically in some patients with Kearns-Sayre and Pearson syndromes, as well as with mutations in *SARS2*.[147-149] Presumably, the impaired energy supply limits magnesium transport, explaining the renal wasting. Why hypomagnesemia is such an inconstant feature in mitochondrial cytopathies remains unclear.

OTHER HYPOMAGNESEMIAS: FAMILIAL HYPOMAGNESEMIA WITH SECONDARY HYPOCALCEMIA

Familial hypomagnesemia with secondary hypocalcemia (HSH) is a rare autosomal recessive disorder caused by loss-of-function mutations in the *TRPM6* gene encoding the TRPM6 channel essential for the intestinal and renal reabsorption of magnesium (see Fig. 73.6).[150] Patients with HSH typically present in infancy or early childhood, with generalized convulsions associated with extremely low levels of serum magnesium, hypocalcemia, and inadequately low PTH levels. Because of the combined intestinal uptake and renal reabsorption defect, improvement of serum magnesium levels is challenging. Despite persistently low levels, however, the prognosis in HSH is favorable with oral supplementation,

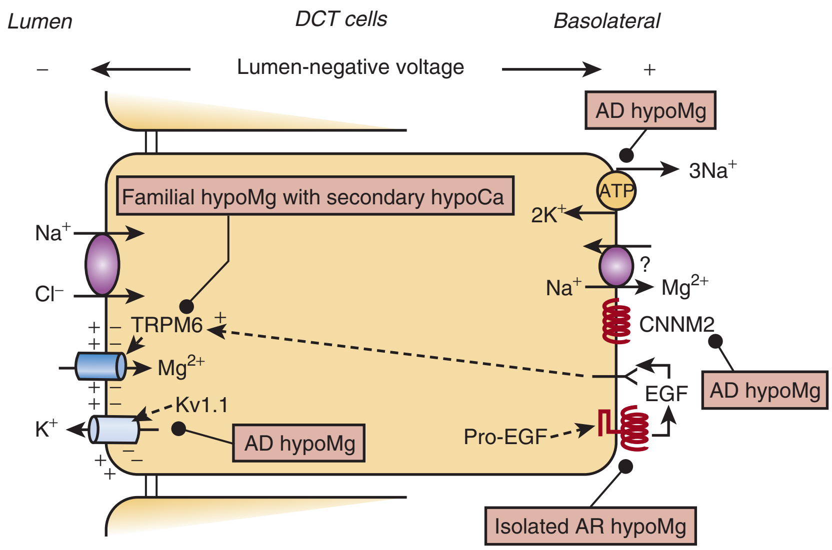

Fig. 73.6 Schematic representation of a cell of the distal convoluted tubule *(DCT)* showing the transport systems involved in the reabsorption of magnesium and its inherited disorders. The thiazide-sensitive Na^+-Cl^- cotransporter, the magnesium-transporting channel TRPM6, and the voltage-gated Kv1.1 potassium channel are depicted on the luminal membrane of the cell. The Na^+-K^+-ATPase, a putative Na^+-Mg^{2+} exchanger, the cyclin M2 protein *(CNNM2)*, the receptor of epidermal growth factor *(EGF)*, and the precursor of this peptide, the membrane protein pro-EGF, are represented on the basolateral side (see text for details). *AD,* Autosomal dominant; *AR,* autosomal recessive; *ATP,* adenosine triphosphate; *ATPase,* adenosine triphosphatase; *hypoCa,* hypocalcemia; *hypoMg,* hypomagnesemia; *TRPM6,* transient receptor potential channel melastatin 6.

although persistent neurologic damage has been described in patients with delayed diagnosis and prolonged seizures.[151]

Hypomagnesemia has been described further with mutations in *CNNM2, KCNA1, EGF,* and *EGFR,* as well as *FAM111A.*[131] All these are exceedingly rare and are described in more detail in Chapters 18 and 44.

CALCIUM AND PHOSPHATE DISORDERS

CALCIUM AND PHOSPHATE CONCENTRATIONS IN SERUM

The normal ranges for calcium and phosphate change during the first few years of life, perhaps reflecting the high requirement for bone formation during this phase of fastest growth. The normal range for total serum calcium is 2.4 to 2.8 mmol/L (9.3–11.1 mg/dL) in a neonate and decreases slowly so that around the age of 7 years the normal range is the same as in adults at 2.2 to 2.6 mmol/L (8.8–10.4 mg/dL).[152] Similarly, the normal range for phosphate in a neonate is 1.4 to 2.3 mmol/L (4.3–7.1 mg/dL) and decreases slowly, so that around the age of 7 years, the normal range is the same as in adults, at 0.9 to 1.6 mmol/L (2.8–5.0 mg/dL).[152]

This neonatal hyperphosphatemia is linked to higher tubular reabsorption of phosphate, potentially due to a blunted tubular response to the already low circulating PTH normally encountered in neonatal life.

VITAMIN D, PARATHYROID HORMONE, CALCITONIN, AND FIBROBLAST GROWTH FACTOR 23

The control of calcium and phosphate homeostasis is complex (Figs. 73.7 and 73.8). For the purposes of this chapter, we

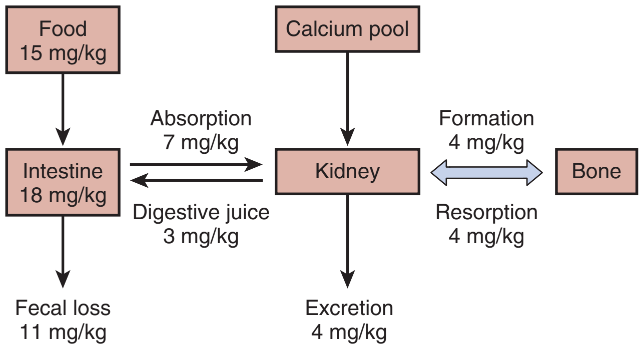

Fig. 73.7 The calcium pool is sustained essentially by balance among dietary intake, intestinal absorption, and renal excretion of calcium. Every day, the skeletal system uses calcium at 4 mg/kg body weight per day to form new bone. This rate of calcium accretion in new bone formation is balanced by the same rate of calcium resorption from bone. Calcium in food is the most important source. Food provides calcium nutrition of 15 mg/kg body weight per day on a North American diet. Approximately 7 mg/kg is absorbed to augment the calcium pool. Digestive juice contributes 3 mg/kg to bring the total intestinal calcium content to 18 mg/kg body weight per day, out of which fecal loss is 11 mg/kg body weight per day. Finally, the renal excretion is 4 mg/kg body weight per day to maintain calcium balance.

shall focus here on four of the key regulators engaged in maintaining calcium and phosphate balance: vitamin D, parathyroid hormone, calcitonin, and fibroblast growth factor 23 (FGF-23; Fig. 73.9).

VITAMIN D

Vitamin D exists in two principal forms: vitamin D_2 (ergocalciferol) and D_3 (cholecalciferol). They result from the ultraviolet (UV) irradiation of ergosterol (D_2), which is mainly found in plants and fungi, or 7-dehydrocholesterol (D_3), which is

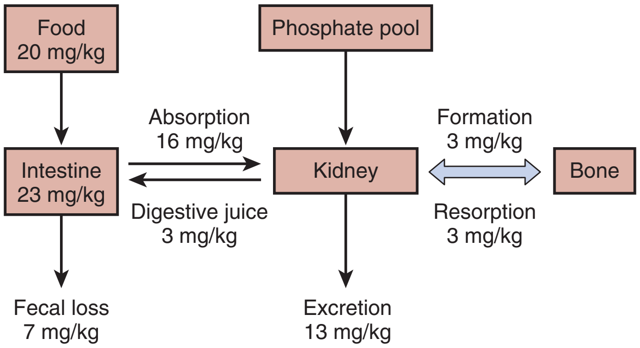

Fig. 73.8 The phosphate pool is sustained essentially by dietary intake, intestinal absorption, and renal excretion of phosphate. Every day, the skeletal system uses phosphate at 3 mg/kg body weight per day to form new bone. This rate of phosphate used in new bone formation is balanced by the same amount of phosphate resorption from bone. Phosphate in food is the most important source. Food provides phosphate nutrition of 20 mg/kg body weight per day on a North American diet. Approximately 16 mg/kg body weight per day is absorbed to augment the phosphate pool. Digestive juice contributes 3 mg/kg body weight per day to bring the total intestinal phosphate content to 23 mg/kg body weight per day, out of which fecal loss is 7 mg/kg body weight per day. Finally, the renal excretion of phosphate is 13 mg/kg body weight per day to maintain phosphate balance.

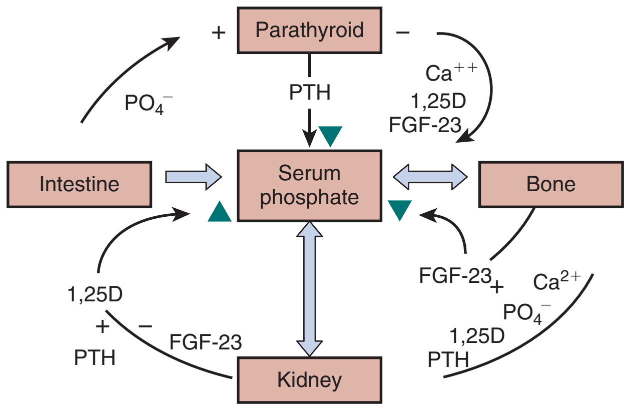

Fig. 73.9 Serum phosphate homeostasis is conserved by three potent hormones, the parathyroid hormone *(PTH)*, fibroblast growth factor 23 (FGF-23), and 1,25-dihydroxyvitamin D (1,25D). As indicated by the *green arrowheads* in the diagram, PTH and FGF-23 lower serum phosphate concentrations and 1,25D increases serum phosphate concentrations. Secretion of these hormones is regulated by reciprocal relationships and by the interaction of calcium *(Ca^{2+})* and phosphate *(PO_4^-)*. Specifically, the kidney's production of 1,25D is stimulated by low serum PO_4^- and PTH and inhibited by FGF-23 levels. At the parathyroid gland, high PO_4^- stimulates the production of PTH, which increases phosphaturia.

from animal sources.[153] Vitamin D is further activated in two hydroxylation reactions; the first one occurs in hepatocytes, which activate it to 25-hydroxyvitamin D (25[OH]D) by the enzymatic activity of CYP2R1 and CYP27A1. The second occurs in the proximal tubule, generating 1,25-dihydrovitamin D (1,25[OH]$_2$D).

Vitamin D in its various forms in blood is mainly bound to vitamin D–binding protein (DBP) and, to a much lower degree (~10%), to albumin. Yet, only unbound vitamin D is bioavailable and can thus bind to the vitamin D receptor.[154] Of note, common polymorphisms affect the levels of DBP, leading to ethnic differences.[155] Because the usual assays for vitamin D measure the total concentration, these can vary, despite similar levels of bioavailable vitamin D. Thus, US blacks have been shown to have lower total vitamin D levels than US whites, yet similar levels for bioavailable vitamin D.[155]

Despite its molecular weight of around 52 kDa, DBP is filtered by the glomerulus and is reabsorbed in the proximal tubular brush border via megalin (LRP2).[156] It is through this process of filtration and reabsorption that vitamin D reaches the mitochondria in the proximal tubule, where the final activation step to 1,25-dihydrovitamin D takes place via the 1α-hydroxylase (CYP27B1). The activity of this key enzyme in vitamin D activation is tightly regulated, stimulated by low concentrations of circulating 1,25(OH)$_2$D, calcium, and phosphate, as well as high concentrations of circulating PTH and inhibited by FGF-23 and Klotho.[157,158]

The key enzyme for 1,25(OH)$_2$D catabolism is CYP24A1. In a five-step progression, starting with 24-hydroxylation, CYP24A1 catabolizes 1,25(OH)$_2$D into water-soluble calcitroic acid.[159]

A high level of 1,25(OH)$_2$D inhibits 1α-hydroxylase and stimulates the production of FGF-23 and Klotho, both of which inhibit 1α-hydroxylase. Such feedback loops precisely control 1,25(OH)$_2$D production.[160]

A high extracellular ionized calcium concentration activates the calcium-sensing receptor (CaSR) in the parathyroid chief cell. CaSR belongs to the family of G protein–coupled receptors, and activation triggers a signaling cascade that suppresses PTH secretion.[161]

CALCITONIN, FIBROBLAST GROWTH FACTOR 23, AND PARATHYROID HORMONE

Calcitonin, produced by the parafollicular cells (C cells) of the thyroid gland, inhibits calcium resorption from osteoclasts. Calcitonin operates via the same CaSR as PTH, but the action of calcitonin is opposite to that of PTH.

FGF-23 from osteocytes is a potent phosphaturic agent. In concert with calcium, phosphate, 1,25(OH)$_2$D, and parathyroid hormone, FGF-23 plays an important role in regulating phosphate homeostasis.[162]

FGF-23 decreases the serum phosphate concentration by downregulating expression of sodium phosphate transporters, both NaPi-IIa and NaPi-IIc, in the proximal tubular membrane. FGF-23 and Klotho inhibit renal 1α-hydroxylase expression. FGF-23 promotes 24-hydroxylase expression.[162] Consequently, 1,25(OH)$_2$D production is reduced, resulting in lowered phosphate absorption from the intestine and bone.

In contrast, with a low serum phosphate concentration and/or low serum PTH concentration, the kidneys increase production of 1,25(OH)$_2$D (see Fig. 73.9). Consequently, the intestinal absorption of phosphate is increased.

At the parathyroid glands, high serum phosphate concentrations stimulate PTH production, promoting phosphaturia to restore serum phosphate levels to normal concentrations. In essence, PTH and FGF-23 reduce the serum phosphate level, whereas 1,25(OH)$_2$D increases it (see Fig. 73.9). These three hormones are regulated by reciprocal interaction and calcium and phosphate levels in serum.

HYPOCALCEMIA

The causes of hypocalcemia are listed in Table 73.11. Mild hypocalcemia is often without apparent symptoms, whereas severe hypocalcemia can present with symptoms of paresthesia, tetany, positive Chvostek and Trousseau signs, prolongation of the QT interval, QRS and ST complex changes, and eventually grand mal seizures and ventricular arrhythmias.

TYPES AND CAUSES

Early Neonatal Hypocalcemia

Early neonatal hypocalcemia, presenting in the first 3 days of life, is frequently seen in premature infants of diabetic mothers and in asphyxiated neonates. The hypocalcemia is due to an inadequate parathyroid response associated with the hypomagnesemia common in diabetic pregnant mothers

Table 73.11 Causes of Hypocalcemia

Type of Hypocalcemia	Causes
Neonatal Hypocalcemia	
Early neonatal hypocalcemia (first few days of life)	Maternal hyperparathyroidism • Maternal diabetes mellitus • Toxemia of pregnancy • Sepsis • SGA, IUGR, prematurity • Asphyxia • Transfusion (citrated blood products) • Congenital rubella • Hypomagnesemia • Respiratory or metabolic alkalosis
Late neonatal hypocalcemia (4th–10th day of life)	Vitamin D deficiency—nutritional deficiency; VDR loss-of-function mutation; deficient 1α-hydroxylase activity Phosphate overload—excessive intake of evaporated or whole milk Nutritional calcium deficiency Hypomagnesemia Hypoalbuminemia (nephrotic syndrome) Transfusion (citrated blood products) Acute or chronic kidney insufficiency Diuretics (furosemide) Organic acidemia Primary hypoparathyroidism—DiGeorge syndrome, familial hypoparathyroidism, pseudohypoparathyroidism, Kenny-Caffey syndrome, partial deletion of GCMB, retardation dysmorphism syndrome, Pearson syndrome, Kerns-Sayre syndrome, PTH gene defects, CaSR-activating gene mutation
Hypocalcemia in Childhood	
Parathyroid-related hypocalcemia	Primary hypoparathyroidism—DiGeorge syndrome, familial hypoparathyroidism, pseudohypoparathyroidism, Kenny-Caffey syndrome, Sanjad-Sakati syndrome, partial deletion of GCMB, retardation dysmorphism syndrome, Pearson syndrome, Kerns-Sayre syndrome, PTH gene defects, CaSR-activating gene mutation Secondary hypoparathyroidism—radiation, surgery, infiltration (hemochromatosis, thalassemia, Wilson disease) Autoimmune polyglandular syndrome type 1 Nutritional vitamin D deficiency • Defective 1α-hydroxylase activity • VDR loss-of-function mutation
Vitamin D–related hypocalcemia	Nutritional calcium deficiency Hypomagnesemia Hyperphosphatemia—kidney failure, rhabdomyolysis, tumor lysis Hypoalbuminemia (nephrotic syndrome) Medications—diuretics, chemotherapeutics, transfusion (citrated blood) Organic acidemia (IVA, MMA, PPA)

CaSR, Calcium-sensing receptor; *GCMB,* glial cell missing homolog B (a parathyroid-specific transcription factor); *IUGR,* intrauterine growth retardation; *IVA,* isovaleric acidemia; *MMA,* methylmalonic acidemia; *PPA,* propionic acidemia; *PTH,* parathyroid hormone; *SGA,* small for gestational age; *VDR,* vitamin D receptor.

and compounded by hyperphosphatemia secondary to delayed phosphaturic action of PTH.[163]

Late Neonatal Hypocalcemia

Late neonatal hypocalcemia presents at 5 to 10 days of life and is commonly due to anticonvulsants administered to the pregnant mother.[163] Phenytoin and phenobarbital anticonvulsants amplify the hepatic clearance of vitamin D, resulting in hypovitaminosis D in the mother and neonate. The consequent impaired absorption of calcium results in the delayed hypocalcemia. Hypocalcemia can also arise from neonatal hyperphosphatemia, which is the result of three compounding factors: (1) the diminished phosphate excretion of the immature neonatal kidneys and the high circulating GH level; (2) the low circulating PTH level at this stage of life; and (3) the high phosphate load in milk.[163]

Maternal hyperparathyroidism also can be associated with late neonatal hypocalcemia in the first few weeks to as late as 1 year of life due to the suppression of fetal PTH production.[164] Similarly, familial hypocalciuric hypercalcemia (FHH) in the mother also suppresses fetal parathyroid secretion.[165]

Infusions of Lipid or Citrate Blood Products

Infusions of lipid or citrate blood products cause hypocalcemia by complexing with ionized calcium.

Alkalosis

Finally, alkalosis, whether respiratory or metabolic, causes hypocalcemia, by increasing the protein binding of calcium.

Most cases of mild neonatal hypocalcemia resolve after 2 to 4 weeks of life. If hypocalcemia and hyperphosphatemia persist beyond this period of time, further workup for hypocalcemia is essential to rule out vitamin D metabolic defects, isolated hypoparathyroidism, or pseudohypoparathyroidism. Finally, neonates with chronic kidney failure are expected to show hypocalcemia, hyperphosphatemia, and elevated concentrations of serum PTH, as well as urea nitrogen and creatinine.

DIAGNOSTIC WORKUP FOR HYPOCALCEMIA

The diagnostic workup for hypocalcemia in the neonate, infant, or child follows the algorithm presented in Fig. 73.10. The initial steps are a determination of the serum albumin level, spot urine sample for the calcium-to-creatinine ratio to

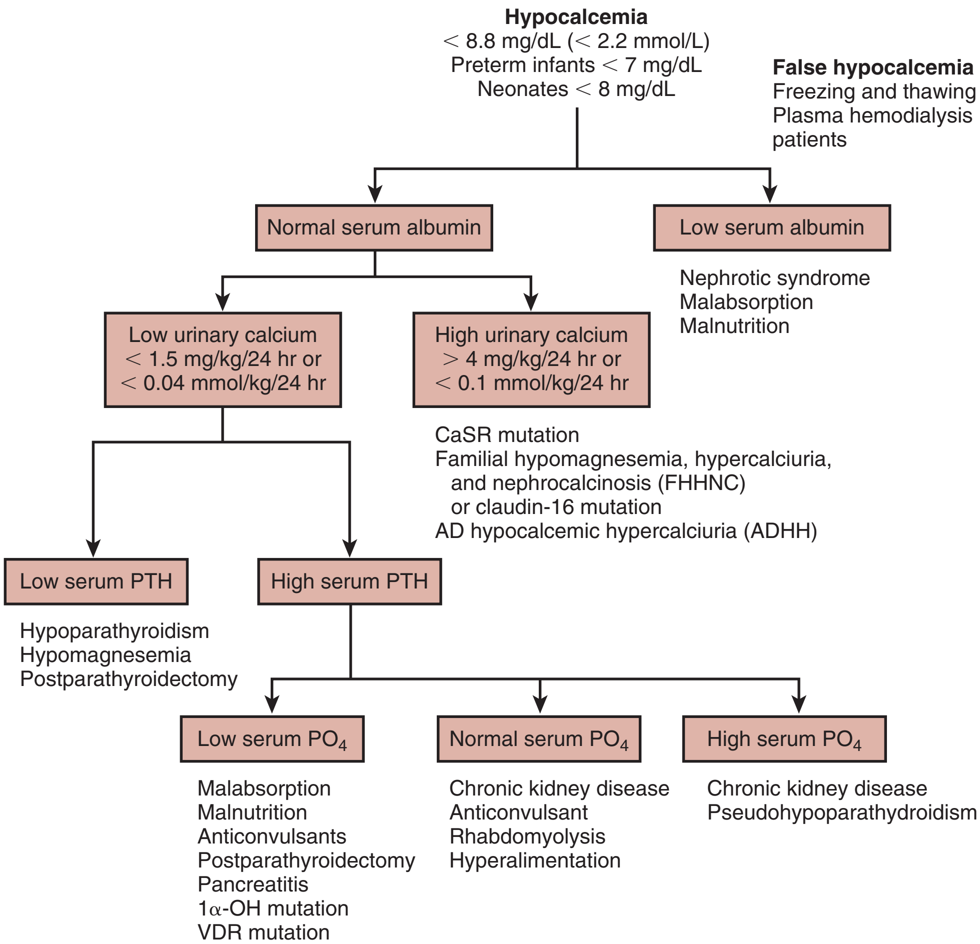

Fig. 73.10 Algorithm for the diagnosis of hypocalcemia. After false hypocalcemia has been excluded by careful history, the first laboratory diagnostic step is to determine serum albumin concentrations. A low serum albumin concentration indicates specific disorders associated with hypocalcemia, such as nephrotic syndrome, malabsorption, and malnutrition. A normal serum albumin concentration indicates an examination of urinary calcium excretion. Low urinary calcium excretion relative to low serum parathyroid hormone *(PTH)* indicates hypoparathyroidism, hypomagnesemia, and postparathyroidectomy. A low urinary calcium excretion associated with high serum PTH relative to low, normal, and high serum PO_4 concentration, respectively, indicates several diagnostic conditions, as listed. *AD,* Autosomal dominant; *CaSR,* calcium-sensing receptor; *1α-OH,* 1α-hydroxylase; *PO_4,* phosphate; *VDR,* vitamin D receptor.

Table 73.12 Treatment of Hypocalcemia

Medication	Dose	Administration
Symptomatic Hypocalcemia in the Neonate		
Calcium gluconate	IV 10% solution, 1 mg/kg (9 mg elemental Ca^{2+}/mL); 1 mL/min over 10 min	Repeat as needed, with careful monitoring.
Magnesium sulfate, 50% solution	0.1 mL/kg IV or IM	
Symptomatic Hypocalcemia in the Infant		
Calcium gluconate	IV 10% solution, 1 mg/kg (9 mg elemental Ca^{2+}/mL), 1 mL/min over 10 min	Repeat as needed, with careful monitoring.
Elemental Ca^{2+} continuous infusion	1 mg elemental Ca^{2+}/kg/hr	
Asymptomatic Hypocalcemia in the Infant		
Calcium carbonate, PO	50 mg elemental calcium/kg/day	Divided into four doses per day

IM, Intramuscularly; *IV,* intravenously; *PO,* orally.

delineate urinary calcium excretion, and then determination of values for serum PTH, magnesium, phosphate, serum creatinine, and 25-hydroxyvitamin D (25[OH]D) levels and other specific tests, as dictated by the underlying disease causing the hypocalcemia (see Table 73.11). In the presence of persistent hypocalcemia and suspected maternal hypoparathyroidism or vitamin D deficiency, the mother should be assessed.

TREATMENT OF HYPOCALCEMIA

Treatment of acute symptomatic hypocalcemia (Table 73.12) consists of 10% calcium gluconate at a dose of 1 to 2 mL/kg body weight (9.3 mg of elemental calcium/mL) IV at a slow drip of less than 1 mL/min over 10 minutes, with electrocardiographic monitoring of the QT interval until symptoms vanish.[166] If tetany or seizures persist, calcium infusion is repeated as necessary until the resolution of severe symptoms. Alternatively, a continuous infusion of elemental calcium, 1 to 3 mg/kg body weight per hour, can be given. Extravasation of calcium-containing fluids causes tissue necrosis, and thus any calcium infusion must be monitored carefully. If the serum calcium concentration fails to rise with this therapy, primary hypomagnesemia must be considered and, if present, treated with magnesium sulfate IV at a dose of 0.1 to 0.2 mL/kg body weight (50% solution) under cardiac monitoring or given intramuscularly.

Hypocalcemic infants with hyperphosphatemia may benefit from low-phosphate formula or breast milk with an overall calcium-to-phosphate ratio of 4:1. Infants with vitamin D deficiency rickets due to maternal hypovitaminosis D respond well to oral vitamin D (1000–2000 U daily for 4 weeks) and elemental calcium (40 mg/kg body weight per day).

For early neonatal hypocalcemia, this treatment brings about normalization in 2 to 3 days, after which the calcium supplementation is reduced by half for 2 days and then discontinued. If prolonged calcium supplementation is required, hypoparathyroidism, pseudohypoparathyroidism, or defective vitamin D metabolism must be considered.

MONITORING HYPOCALCEMIA

Monitoring of hypocalcemia during treatment includes determination of fasting total and ionized serum calcium concentration, fasting serum phosphate concentration, and spot urine examination for the calcium-to-creatinine ratio. A 24-hour urine collection at home before the clinic visit is recommended to measure calcium and creatinine excretion in continent children.

The goal is to keep 24-hour urinary calcium excretion below 4 mg (0.1 mmol)/kg body weight per day to prevent hypercalciuria and nephrocalcinosis related to the calcium supplementations and vitamin D administration. The serum PTH level is included in the monitoring if hypocalcemia is not due to PTH deficiency or resistance. Serum magnesium concentrations should be checked at least annually.

HYPERCALCEMIA

Although the range of normal for serum calcium concentrations varies with age and among laboratories, a serum calcium concentration of 12 mg/dL (3.0 mmol/L) is usually considered mild hypercalcemia, between 12 and 14 mg/dL (3.0–3.5 mmol/L) it is considered moderate, and above 14 mg/dL (3.5 mmol/L) it is considered severe and is typically associated with symptoms.[167]

Presenting features of hypercalcemia include hypertension and GI symptoms due to its effect on increasing the contractility of cardiac and smooth muscle. Complaints of weakness, hypotonia, anorexia, nausea, colic, and constipation are common. The polyuria and dehydration are caused by hypercalcemia inhibiting the action of AVP on the collecting tubules. The vasoconstriction of hypercalcemia reduces renal blood flow and the GFR, with the fall in GFR aggravated by the dehydration. Nephrocalcinosis ensues from prolonged hypercalcemia and dehydration. Acute pancreatitis associated with hypercalcemia is related to microcalcinosis. How hypercalcemia induces somnolence, lethargy, stupor, and coma is poorly understood. Vasoconstriction reducing cerebral blood flow is considered a possible mechanism.

Table 73.13 lists the causes of hypercalcemia. This list is extensive, and many of the individual conditions are covered elsewhere in this text. Here we will focus on the following four conditions—idiopathic infantile hypercalcemia, Williams-Beuren syndrome, familial hypocalciuric hypercalcemia, and neonatal severe hyperparathyroidism.

Table 73.13 Causes of Hypercalcemia

Type	Cause
Neonatal hypercalcemia	Maternal hypervitaminosis D Idiopathic infantile hypercalcemia Excess calcium and vitamin D intakes Infantile hypothyroidism • Williams-Beuren syndrome
PTH-dependent hypercalcemia	Primary hyperparathyroidism—adenoma, hyperplasia, carcinoma Tertiary hyperparathyroidism Drug-induced: lithium Familial hypocalciuric hypercalcemia PTH receptor gain-of-function mutation (Jansen syndrome) Autoimmune hypocalciuric hypercalcemia
PTH-independent hypercalcemia	Immobilization Vitamin D excess—ingestion or topical analogues, granulomatous disease, Williams-Beuren syndrome Drug induced—thiazides, milk-alkali syndrome, estrogen, vitamin A intoxication, aminophylline Kidney insufficiency—acute kidney injury, chronic kidney disease with aplastic bone disease Neoplasm—multiple endocrine neoplasia types 1 and 2, lytic bone metastases, hyperparathyroidism–jaw tumor syndrome PTH receptor excess (nonneoplastic) Thyrotoxicosis • Adrenal insufficiency

PTH, Parathyroid hormone.

IDIOPATHIC INFANTILE HYPERCALCEMIA

In Britain in the 1950s, vitamin D supplementation of infant formula and fortified milk at high doses (up to 4000 IU/day) brought about an epidemic of infantile hypercalcemia, presenting with failure to thrive, intermittent fever, vomiting, dehydration, and nephrocalcinosis. A number of fatalities occurred. Vitamin D hypersensitivity and/or defective catabolism of vitamin D was suspected, but never resolved. In response to these fatalities, a dramatic reduction in vitamin D supplementation to 400 IU/day was enforced, resulting in a significant drop in idiopathic infantile hypercalcemia in Britain.

Later evidence showed loss of function of *CYP24A1* in some of these patients, causing poor degradation of $1,25(OH)_2D$.[159] This was the first gene identified associated with autosomal recessive idiopathic infantile hypercalcemia. Subsequently and somewhat surprisingly, mutations in the gene *SLC34A1*, encoding NaPI-IIa, were identified as another cause.[168] Initial data have indicated that the primary renal phosphate wasting, with consequent hypophosphatemia, induces inappropriate 1-alpha hydroxylation of cholecalciferol, leading to hypercalcemia. Importantly, data from some patients have indicated that the hypercalcemia can be ameliorated by phosphate supplementation.[168] Thus, identification of *SLC34A1*-asscoiated infantile hypercalcemia, either by low plasma phosphate levels or by genetic testing, may have important therapeutic consequences.

WILLIAMS-BEUREN SYNDROME

Microdeletion at chromosome 7q11.23 gives rise to the multisystem disorder known as Williams-Beuren syndrome (OMIM no. 194050; also known as Williams syndrome), which is characterized by hypercalcemia, supravalvular aortic and pulmonary stenosis, growth retardation, and variable gradations of mental retardation.[169]

Williams-Beuren syndrome (prevalence, 1/10,000) is a contiguous gene syndrome, which may reveal genetic factors underlying calcium and cardiovascular lesions, hypertension, glucose intolerance, and anxiety disorders. A critical region at 7q11.23 for this syndrome is a microdeletion of the elastin gene *(ELN)*. Interestingly, Williams-Beuren syndrome is not caused by point mutations in *ELN*. Instead, a segment of an allele of the *ELN* gene is completely lacking. The fluorescence in situ hybridization (FISH) test includes *ELN*-specific probes and has become the most widely used laboratory test to establish the diagnosis of Williams-Beuren syndrome.

Presenting features include round, elfin face and full lips, broad forehead, strabismus, flat nasal bridge, upturned nostrils, dental malocclusion, hypodontia, early graying hair, and slight premature aging. Short stature and obesity are persistent features. Hypotonia, poor balance, and reduced coordination, scoliosis, lordosis, joint laxity, and hyperreflexia, and inguinal hernia and rectal prolapse are findings described in patients with Williams-Beuren syndrome. These patients are often friendly, engaging, and endearing, with a high musical ability. Cognitive impairment varies over a wide range, with IQ scores from 40 to 100 (mean IQ, 55). Some patients suffer from hypersensitivity to sound, visuospatial weaknesses, and enormous learning difficulties, but selective language skills.

An episode of hypercalcemia is seen in 5% to 50% of cases and is usually mild but can be more severe, especially during infancy.[169] Transient hypercalciuria usually accompanies the duration of the hypercalcemic episode. Whether the hypercalcemic episode is the result of vitamin D hypersensitivity as originally proposed or of elevated $1,25(OH)_2D$ levels or defective calcitonin production has not been resolved.

FAMILIAL HYPOCALCIURIC HYPERCALCEMIA

FHH is a rare, typically asymptomatic autosomal dominant condition, with mild to moderate elevation of total and ionized calcium concentration. Serum vitamin D concentrations are normal. Serum PTH concentrations are normal in most patients, but in 15% to 20% of patients, the PTH level is mildly elevated. Serum magnesium concentrations are normal or slightly elevated. Serum phosphate concentrations are low or normal. Bone mineral density is normal.

FHH has been regarded as a benign, albeit lifelong condition, with a small risk for pancreatitis and chondrocalcinosis. Hence, FHH is also known as *familial benign hypocalciuric hypercalcemia*.

Three variants of FHH have been identified, based on genetics, providing insights into the signaling pathways of CaSR[170]:

1. Type 1 FHH is due to loss-of-function mutations of CaSR, a guanyl nucleotide–binding protein (G protein)–coupled

receptor.[171] CaSR signals by means of a G protein subunit α11 ($G_{\alpha 11}$).

2. Type 2 FHH is due to loss-of-function mutations of $G_{\alpha 11}$, encoded by the *GNA11* gene, which colocalizes with the FHH type 2 loci on chromosome 19p13.3.[172]
3. Type 3 FHH is due to mutations of adaptor-related protein complex 2, σ_1-subunit (AP2S1), which modifies CaSR endocytosis.[173] Patients with this form typically have more severe hypercalcemia than in the other two forms, and some have additional features, such as intellectual impairment, consistent with the relevance of AP2S1 beyond calcium regulation.

The hypercalcemia of FHH is usually asymptomatic and does not need medical intervention. Occasionally a calcimimetic agent—for example cinacalcet—has been used successfully.

NEONATAL SEVERE HYPERPARATHYROIDISM

Neonatal severe hyperparathyroidism is an extremely rare autosomal recessive disorder, characterized by relentlessly symptomatic hypercalcemia, elevated PTH level, severe parathyroid hyperplasia, and hyperparathyroid bone disease. It is caused by the recessive inheritance of loss-of-function mutations in the *CASR* gene. The hypercalcemia of neonatal severe hyperparathyroidism is fatal without a total parathyroidectomy. Postoperatively, the patient needs lifelong vitamin D and calcium supplementation.

DIAGNOSTIC WORKUP FOR HYPERCALCEMIA

The diagnostic workup for hypercalcemia follows the algorithm presented in Fig. 73.11. It starts by ruling out false hypercalcemia by careful history to exclude laboratory error and phlebotomy-induced hemolysis. Further investigations include PTH and a spot urine test to determine the calcium-to-creatinine ratio. In the context of a low serum PTH concentration, examination of PTH-related protein (PTHrP), followed by serum $25(OH)D_3$ and $1,25(OH)_2D_3$, is indicated. It is essential to examine the family history for FHH, as shown in the algorithm; search for occult neoplasms, including multiple endocrine neoplasia syndromes; and review the history of medications (see Table 73.13) with side effects of hypercalcemia.

TREATMENT OF HYPERCALCEMIA

Treatment of patients with symptomatic hypercalcemia (Table 73.14) consists of rehydration with normal saline to restore

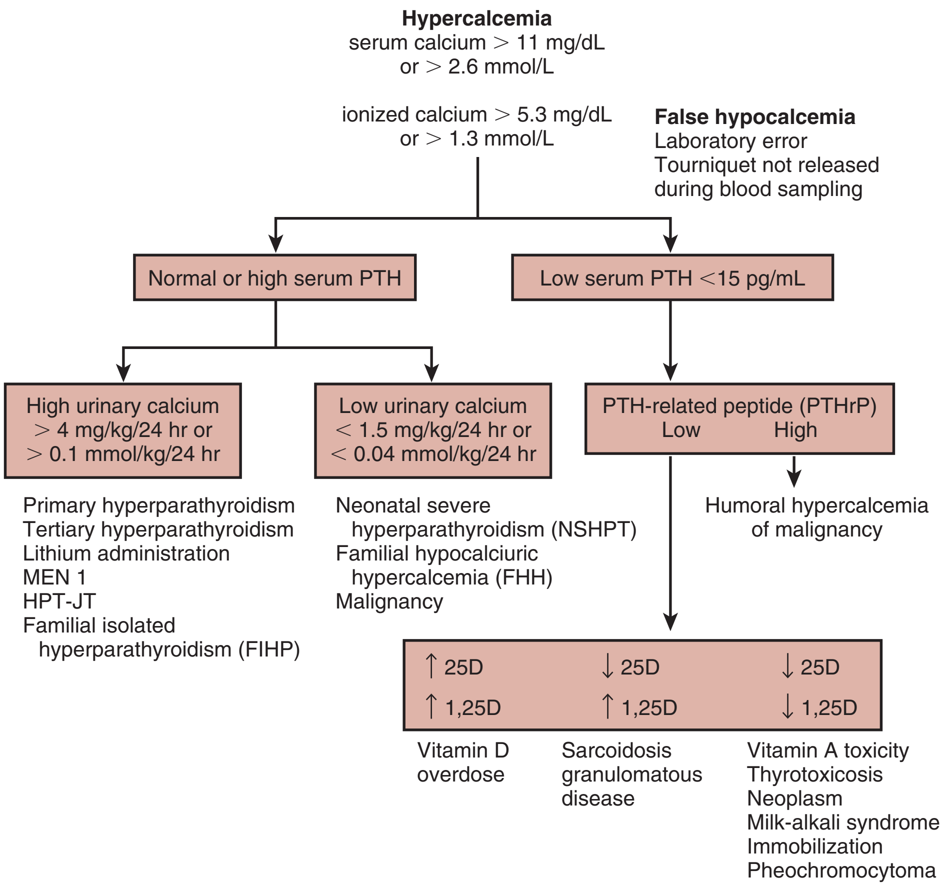

Fig. 73.11 Algorithm for the diagnosis of hypercalcemia. After false hypercalcemia has been excluded by careful history, the first laboratory diagnostic step is to determine serum parathyroid hormone *(PTH)* concentrations. A low serum PTH level in relation to PTH-related peptide *(PTHrP)* suggests specific disorders associated with hypercalcemia and may need an examination of vitamin D metabolites—25 hydroxyvitamin D (25D) and 1,25-dihydroxyvitamin D (1,25D)—for differential diagnosis. A normal or high serum PTH concentration calls for an examination of urinary calcium excretion. High urinary calcium excretion relative to normal or high serum PTH concentration indicates several diagnostic considerations. Low urinary calcium excretion with a normal or high serum PTH concentration yields another set of diagnostic conditions, as listed in the algorithm. *HPT-JT,* Hyperparathyroidism–jaw tumor syndrome; *MEN,* multiple endocrine neoplasia.

Table 73.14 Treatment Options for Symptomatic Hypercalcemia

Therapy	Dosage	Administration
Saline infusion	3000 mL/m^2/day IV	This is twice the daily maintenance; reduce to maintenance dose as soon as serum calcium level is <12 mg/dL.
Furosemide	1 mg/kg IV	q12–24 hr
Severe hypercalcemia (>14 mg/dL or 3.5 mmol/L)		
• Bisphosphonate (e.g., pamidronate)	1–2 mg/kg IV over 4 hr	Once
• Calcitonin	4–8 IU/kg SC	q12–24 hr
• Dialysis		

IV, Intravenously; *q*, every; *SC,* subcutaneously.

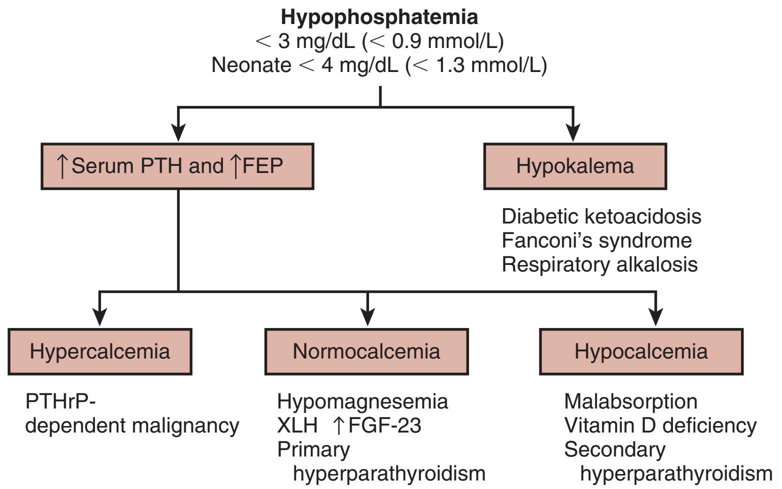

Fig. 73.12 Algorithm for the diagnosis of hypophosphatemia. The diagnostic workup requires determination of the serum parathyroid hormone *(PTH)* level and fractional excretion of phosphate *(FEP)*, as well as serum potassium concentrations. Hypophosphatemia in the presence of increased serum PTH levels, and FEP requires a determination of the serum calcium concentrations. *FGF-23,* Serum fibroblast growth factor 23; *PTHrP,* PTH-related peptide; *XLH,* X-linked hypophosphatemia.

the normal circulating volume and improve the GFR and calcium excretion.[174] To further augment calcium excretion, furosemide can be given once the patient is euvolemic. To keep up with the furosemide-induced diuresis, additional fluid intake must be provided. Once the calcium concentration falls below 12 mg/dL (3.0 mmol/L), furosemide is no longer needed.

Severe hypercalcemia with serum values exceeding 14.0 mg/dL (3.5 mmol/L) can be treated with bisphosphonate to impair bone reabsorption. To prevent recurrent dehydration during treatment, monitoring of fluid intake and output is critical. In addition, dialysis to remove calcium has been effective. Finally, parathyroidectomy may be indicated in cases of severe hypercalcemia from primary hyperparathyroidism that fails to respond to medical therapy.

HYPOPHOSPHATEMIA

A serum phosphate concentration of 1.5 to 3.5 mg/dL (0.48–1.12 mmol/L) is typically considered mild hypophosphatemia and asymptomatic. A serum phosphate concentration less than 1.5 mg/dL (0.48 mmol/L) in a child is severe and usually symptomatic. Clinical symptoms are related to a reduction in intracellular adenosine triphosphate level, increasing the risks for hemolysis, rhabdomyolysis, and myopathies including respiratory and cardiac failures. Paradoxically, intracellular phosphate released from cell breakdown may result in normophosphatemia or hyperphosphatemia, which may mask the true hypophosphatemia.

The causes of hypophosphatemia are listed in Table 73.15.

DIAGNOSTIC WORKUP FOR HYPOPHOSPHATEMIA

The diagnostic workup for hypophosphatemia follows the algorithm presented in Fig. 73.12 and starts with the serum PTH concentration and urine for the determination of renal phosphate handling. The tubular maximum reabsorption of phosphate (TmP/GFR) is calculated as follows:

$$\text{TmP/GFR} = P_{\text{phosphate}} - (U_{\text{phosphate}} \times P_{\text{creatinine}}/U_{\text{creatinine}})$$

These values are all in consistent units, mmol/L or mg/dL. Age-dependent normal ranges for TmP/GFR are detailed in Table 73.16. Alternatively, the tubular reabsorption of phosphorus (TRP) can be calculated, which is the complement to the fractional excretion of phosphorus (FEP)—that is, TRP (in %) = 100 − FEP (in %). Typically, a TRP greater than 85% is considered normal, but this can be falsely normal if the serum phosphate level is so low that the impaired tubular reabsorption can cope with the decreased filtered

Table 73.15 Causes of Hypophosphatemia

Cause	Features
Decreased phosphate intake	• Starvation, inadequate phosphate intake, chronic diarrhea • Total parenteral nutrition with insufficient phosphate content
Increased loss of phosphate	Increased renal phosphate excretion • Primary hyperparathyroidism • Secondary hyperparathyroidism—vitamin D deficiency or resistance (including 1α-hydroxylase deficiency, VDR mutations, VDDR); imatinib • Excess FGF-23 or phosphatonins, X-linked hypophosphatemia, AD hypophosphatemic rickets, tumor-induced osteomalacia, epidermal nevus, McCune-Albright syndrome • Fanconi syndrome, cystinosis, Wilson disease, Dent disease, Lowe syndrome, multiple myeloma, amyloidosis, heavy metal toxicity, rewarming of hyperthermia, Na/Pi-IIa and Na/Pi-IIc mutation (HHRH) • PTHrP-dependent hypercalcemia of malignancy • Hypomagnesemia Decreased intestinal absorption • Vitamin D deficiency or resistance (VDDR types 1 and 2) • Malabsorption Increased intestinal loss • Phosphate binding antacids used in treating peptic ulcers Increased losses from other routes • Skin—severe burns • Vomiting
Phosphate shifting from extracellular compartment to cells and bones	Diabetic ketoacidosis • Alcohol intoxication • Acute respiratory alkalosis, salicylate intoxication, gram-negative sepsis, toxic shock syndrome, acute gout • Refeeding syndromes from starvation, anorexia nervosa, hepatic failure: acute intravenous glucose, fructose, glycerol • Rapid cellular proliferation—intensive erythropoietin therapy, GM-CSF therapy, leukemic blast crisis • Recovery from hypothermia • Heat stroke • Postparathyroidectomy; “hungry bone” disease—osteoblastic metastases, antiresorptive treatment of severe Paget disease • Catecholamines (e.g., albuterol, dopamine, terbutaline, epinephrine) • Thyrotoxic periodic paralysis • Hypocalcemic periodic paralysis
Miscellaneous	• Hyperaldosteronism • Oncogenic hypophosphatemia • Post–kidney transplantation • Post–partial hepatectomy • High-dose corticosteroids, estrogens • Medications—ifosfamide, toluene, calcitonin, bisphosphonate, tenofovir, paraquat, cisplatin, acetazolamide and other diuretics • Postobstructive diuresis

AD, Autosomal dominant; *FGF-23,* fibroblast growth factor 23; *GM-CSF,* granulocyte-macrophage colony-stimulating factor; *HHRH,* hereditary hypophosphatemic rickets with hypercalciuria; *Na/Pi-II,* type II sodium-dependent phosphate cotransporter; *PTHrP,* parathyroid hormone–related peptide; *VDR,* vitamin D receptor; *VDDR,* vitamin D–dependent rickets.

load of phosphate. For this reason, Tmp/GFR is more reliable because it corrects for the filtered load.[175]

The serum calcium concentration is particularly helpful in the workup for hypophosphatemia. Because PTH is a powerful phosphaturic hormone, a low Tmp/GFR accompanied by hypercalcemia points to primary hyperparathyroidism as the cause of the hypophosphatemia. In contrast, if a low Tmp/GFR is accompanied by hypocalcemia, vitamin D deficiency and malabsorption or malnutrition must be considered.

IMAGING STUDIES IN CHRONIC HYPOPHOSPHATEMIA

Imaging studies in chronic hypophosphatemia may include a skeletal x-ray examination to delineate osteopenia, osteomalacia, and rickets. Ultrasonography of the neck can be helpful if a parathyroid adenoma is suspected. Finally, a scan using technetium 99m–labeled sestamibi as an imaging agent may be indicated to delineate the presence of an ectopic parathyroid gland.

Table 73.16 Age-Appropriate Reference Ranges for Tubular Maximum Reabsorption of Phosphate

Age	Range (mmol/L)	Range (mg/dL)
<1 mo	1.48–3.43	4.58–10.62
1–3 mo	1.48–3.30	4.58–10.22
4–6 mo	1.15–2.60	3.56–8.05
7 mo–2 yr	1.10–2.70	3.41–8.36
2–4 yr	1.04–2.79	3.22–8.64
4 -6 yr	1.05–2.60	3.25–8.05
6–8 yr	1.26–2.35	3.90–7.28
8–10 yr	1.10–2.31	3.41–7.15
10–12 yr	1.15–2.58	3.56–7.99
12–15 yr	1.18–2.09	3.65–6.47
>15 yr	0.80–1.35	2.48–4.18

Modified from Kruse, K, Kracht, U, Gopfert, G: Renal threshold phosphate concentration (TmPO4/GFR). *Arch Dis Child.* 1982;57:217-223.

Table 73.17 Treatment of Severe Hypophosphatemia

Therapy	Dosage	Route of Administration
Asymptomatic Severe Hypophosphatemia		
Elemental phosphate	2.5 mg (0.1 mmol)/kg body weight	IV
Symptomatic Severe Hypophosphatemia		
Elemental phosphate	5 mg (0.2 mmol)/kg body weight	IV over 6 hr

IV (intravenous) treatment is recommended for ventilated patients. In nonventilated asymptomatic patients, oral supplementation with 5–15 mg (0.2–0.6 mmol)/kg/day may suffice.

TREATMENT OF HYPOPHOSPHATEMIA

The treatment of hypophosphatemia from a long-standing condition, such as starvation or diabetic ketoacidosis, is complicated by the risk for rapid progression to severe, life-threatening hypophosphatemia during refeeding, as precipitated by rehydration and glucose or insulin provision. Thus, the concurrent use of phosphate supplementation (Table 73.17) is indicated.

HYPERPHOSPHATEMIA

As discussed earlier in this chapter, the normal range for serum phosphate changes with age, and hyperphosphatemia thus refers to a serum concentration above the age-appropriate reference range. The clinical features of hyperphosphatemia are related to hypocalcemia. In view of the reverse interrelationship between these divalent ions, a high serum phosphate concentration is chemically linked to a low serum ionized calcium concentration, and vice versa.

Table 73.18 Causes of Hyperphosphatemia

Cause	Features
Impaired renal excretion of phosphate	Renal insufficiency • Hypoparathyroidism, pseudohypoparathyroidism • Transient parathyroid resistance of infancy • Acromegaly • Tumoral calcinosis Hyperthyroidism • Juvenile hypogonadism • High ambient temperature • Heparin • Bisphosphonate etidronate
Increased phosphate intake • Exogenous loads • Endogenous loads	• Phosphate salts—laxatives and enemas • Vitamin D intoxication • Blood transfusion • White phosphorus burns • Liposomal amphotericin B • Fosphenytoin • Parenteral phosphate • Crush injury • Rhabdomyolysis • Cytotoxic therapy of neoplasms—tumor lysis • Hemolysis • Malignant hyperthermia • Catabolic states • Lactic acidosis • Fulminant hepatitis
Transcellular shift of phosphate	Cellular shift in diabetes ketoacidosis Metabolic acidosis Respiratory acidosis
Miscellaneous	Hyperostosis

Thus, hyperphosphatemia becomes symptomatic from the hypocalcemia (e.g., tetany and other symptoms, discussed earlier in the section on hypocalcemia). Table 73.18 lists the causes of hyperphosphatemia.

DIAGNOSTIC WORKUP FOR HYPERPHOSPHATEMIA

The diagnostic workup for hyperphosphatemia follows the algorithm presented in Fig. 73.13 and includes serum calcium, PTH, potassium, creatinine, and 25(OH)D concentrations. The suspected underlying disease (see Table 73.18) causing hyperphosphatemia may require other specific tests. Finally, spurious hyperphosphatemia can result from interference with the phosphate measurement by hyperlipidemia, hyperbilirubinemia, and hyperglobulinemia, requiring specific tests for each condition.[176]

IMAGING STUDIES IN HYPERPHOSPHATEMIA

Imaging studies in hyperphosphatemia depend on the underlying disease—for example, wrist x-ray examination to delineate renal osteodystrophy and magnetic resonance imaging to rule out pituitary adenoma in acromegaly.

TREATMENT OF HYPERPHOSPHATEMIA

The treatment of hyperphosphatemia in the context of normal kidney function includes adequate fluid intake to

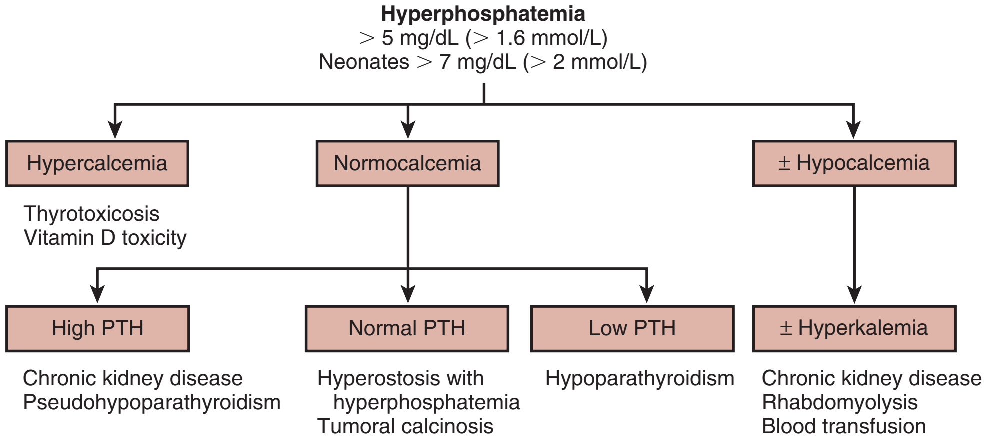

Fig. 73.13 Algorithm for the diagnosis of hyperphosphatemia. The first step is to differentiate between conditions associated with hypercalcemia, normocalcemia, and hypocalcemia, respectively. Subsequently, serum parathyroid hormone *(PTH)* and serum potassium concentrations indicate related diagnoses.

promote diuresis and restriction of phosphate intake with the addition of phosphate binders. Finally, specific treatment for the underlying disease is discussed in other chapters in this text.

SUMMARY AND CONCLUSION

We have highlighted the causes, clinical characteristics, and management of acid-base, water, and electrolyte disorders in pediatric patients. Although the principles of fluid, electrolyte, and acid-base homeostasis are the same as for adults, the specific requirements of the growing child lead to altered age-specific normal ranges (e.g., for calcium, phosphate) and occasionally altered tubular function, such as the impaired urine-concentrating ability during the first year of life. Moreover, the differential diagnosis of homeostatic abnormalities in children is different and much more likely to include mendelian disorders, rather than acquired diseases with secondary electrolyte abnormalities.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Which of the following statements about hyponatremia is true?
   a. Hyponatremia generally reflects a deficiency in salt.
   b. Hyponatremia is always best treated with salt supplementation.
   c. Urine sodium concentration can help distinguish between salt wasting and water overload.
   d. Urine osmolality can help distinguish between water retention and water intoxication.

   Answer: d

   Rationale: Hyponatremia almost always reflects an excess in water than a deficiency in salt. Consequently, it is usually best treated by fluid restriction/removal rather than salt supplementation. Since water overload is associated with hypervolemia, the appropriate physioklogiucal response of the kidneys is to excrete sodium to maintain volume homeostasis. Thus urinary sodium cannot reliably distinguish between salt wasting and water overload. Hyponatremia due to water retention is marked by an inappropriate elevated urine osmolality (>100 mOsm/kg), whereas in pure water intoxication, the kidneys function normally and thus excrete a dilute urine (<100 mOsm/kg).

2. Which of the following statements about renal hypokalemia is incorrect?
   a. Hypokalemia in Gitelman syndrome reflects secondary activation of the renin-angiotensin system (RAS) to compensate for salt loss. Thus salt supplementation stabilises plasma potassium levels.
   b. Hypokalemia can be associated with salt wasting and salt retaining disorders.
   c. In Bartter and Gitelman syndromes, sustained normal plasma potassium levels can usually be achieved with oral supplementation.
   d. In Bartter syndrome, non steroidal anti-inflammatory drugs are used to suppress renin production.

   Answer: c

   Rationale: In contrast to Bartter syndrome, in Gitelman syndrome tubuloglomerular feedback (TGF) is intact and thus renin production in the juxtaglomerular apparatus (JGA) is volume dependent. Stabilisation of extracellular volume by salt supplementation thus decreases RAS activation and stabilises plasma potassium levels. Renin activation in the JGA is mediated by prostaglandins, produced predominantly by the enzyme COX-2, which is inhibited by NSAIDs. TGF is impaired in Bartter syndrome with pathological upregulation of COX-2. Renal hypokalemia is caused by increased activity of potassium secretion in the collecting duct. Potassium secretion is coupled to soidum reabsorption through ENaC and this can be primary (e.g., Liddle syndrome), i.e., salt retaining or secondary to salt wasting (e.g., Bartter and Gitelman syndromes).

3. Which of the following statements is true?
   a. PHA1 and PHA2 are both characterised by hyperkalemia and salt wasting.
   b. Thiazides can normalise all abnormalities associated with PHA1.
   c. Potassium exchange resins can be used to stabilise potassium levels in PHA1.
   d. Hyperkalemia always reflects either potassium retention or excess intake.

   Answer: d

   Rationale: While PHA1 and 2 are both characterised by hyperkalemic acidosis and unresponsiveness to mineralocorticoids, PHA1 is a salt wasting disorder and PHA2 a salt retaining disorder, characterised by hypertension. PHA2 is due to enhanced salt reabsorption in the distal convoluted tubule and thus can be effectively treated with thiazides. Exchange resins can be very helpful especilly in little children with PHA1 to stably suppress plasma potassium levels. Hyperkalemia can be due to excess potassium, but also do to transcellular shift from intracellular to extracellular).

4. Which of the following statements is incorrect?
   a. Isolated proximal RTA (without other rennal tubular abnormalities) is exceedingly rare and typically associated with eye abnormalities.
   b. Proximal RTA occurs most commonly in the context of generalised proximal tubular dysfunction (renal Fanconi syndrome).
   c. The hypercalciuria in distal RTA is due to an associated defect in tubular calcium excretion.
   d. Distal RTA can be effectively treated with alkali supplementation.

   Answer: c

   Rationale: Proximal RTA almost always occurs in the context of a generalised proximal tubular dysfunction. The hypercalciuria in dRTA reflects primarily the buffering of acid by bone with consequent release of calcium into the blood stream. Adequate supplementation of alkali can normalise the acidosis and hypercalciuria of dRTA.